Mechanical Systems and Signal Processing 238 (2025) 113142

Contents lists available at ScienceDirect

Mechanical Systems and Signal Processing

journal homepage: www.elsevier.com/locate/ymssp

Full length article

Observability of nonlinear discrete-time systems with unmeasured inputs☆

M.N. Chatzis [a,*], K. Maes [b], G. Lombaert [b]

[a] The University of Oxford, Department of Engineering Science, Oxford, United Kingdom
[b] KU Leuven, Department of Civil Engineering, Leuven, Belgium

ARTICLE INFO

Communicated by D. Bernal

Keywords:
System identification
Unmeasured inputs
Input estimation
Observability

ABSTRACT

The observability analysis of a dynamic system for a given sensor setup offers useful insights on the ability to estimate its states and parameters. Recent works have further extended the concept of observability to systems under unmeasured inputs, while efficient algorithms have allowed the study of realistic dynamic systems which typically involve multiple degrees of freedom and parameters to be identified. Observability algorithms however typically rely on the assumption that the measurements are obtained continuously in time. While the results obtained from this assumption are still useful, some of the observations from the observability analysis, e.g., the derivatives of the measurements needed for the system to become observable, are hard to link to system identification algorithms. Such algorithms use measurements that have been logged at some finite sampling frequency, while the inputs are often represented using a discretization in time. In this work, the concept of (state, parameter and input) observability under the presence of unmeasured inputs is extended to discrete-time systems. This allows accounting for the measurements and inputs being sampled at some finite frequency. The outcomes of the observability analysis are extended to include the number of measurements needed for a state, parameter or input to become observable. The work will discuss how this property is especially important in the case of unmeasured inputs. An efficient algorithm to calculate the observability of discrete systems under the presence of unmeasured inputs is introduced. An efficient numerical implementation of the algorithm is introduced using stabilization of the results across different numerical precisions. The algorithm is then used for dynamic systems which are discretized by methods of different order, and the effect of this process on the observability properties of the system is discussed. Suitable examples demonstrate how the observability of discrete-time systems with unmeasured inputs can guide significant decisions in terms of sensor placement and the modelling of the system.

1. Introduction

The use of sensors to monitor dynamic systems offers a wealth of information that can be used to reduce the uncertainty of the system through estimating its properties and states, i.e., System Identification [1], or inferring the response of the system at critical unmeasured locations and assessing its condition, i.e., Structural Health Monitoring [2].

Substantial improvement of the corresponding algorithms has been achieved over the last decades. A significant development is the relaxation of the assumptions related to the excitation of the system [3]. In the past, the excitation was typically either assumed

☆ Postprint submitted to Mechanical Systems and Signal Processing.

* Corresponding author.
E-mail address: manolis.chatzis@eng.ox.ac.uk (M.N. Chatzis).

https://doi.org/10.1016/j.ymssp.2025.113142
Received 7 January 2025; Received in revised form 17 June 2025; Accepted 21 July 2025
Available online 13 August 2025
0888-3270/

to be measured, as in the case of input–output methods, or white noise in nature, as in the case of output-only methods. Recent algorithms have allowed for estimating the inputs of the system together with its states and parameters [4,5].The term ‘input’ refers to any exogenous time-varying function. This allows for investigating systems with unmeasured loads, time-varying properties, or unknown interface forces between subsystems. In practice, rather than identifying a continuous-time function for the inputs, the input is represented as vector of values obtained at discrete times through sampling, e.g., with Zero-Order-Hold (ZOH). It should further be noticed that system identification algorithms overall operate on some discrete-time representation of the measurements, inputs and the system.

Similarly to system identification algorithms, observability algorithms have relied on the assumption of the inputs to the system being measured or known [6–8]. Improvements have been made over the last years allowing for the use of the algorithms to larger systems [9–11]. Despite those developments, the assumption of the inputs being known was a limiting factor in the applicability of such algorithms.

Hence following the developments in the field of system identification, observability algorithms for systems with unmeasured inputs have also been developed in the last decade. The Observability Rank Condition (ORC) to systems with unmeasured inputs [12], including the case of systems with direct feedback [13]. Efficient algorithms for the case of rational analytical systems were suggested in [14], while [15] has focused on systems with inputs that can be expressed as polynomial functions of time. An algorithm that includes a termination criterion for the maximum number of steps that need to be investigated before concluding on parameter, state or input being unobservable was suggested in [16]. More recent works have focused on providing means to express the Lie symmetries of systems with unmeasured inputs [17], i.e., the transformations of the states, parameters and unmeasured inputs under which the mapping from measured inputs to measured outputs remains invariant.

In all previous cases, the systems studied were continuous, and as is typically the case the outputs were assumed to be measured continuously. While the conclusions from such studies are useful, the relevant algorithms do not allow for discussing aspects related to how the discretization of the dynamic system or the parametrization of the input affects the observability properties. Yet both are inevitable in the relevant system identification algorithms for input estimation.

There is a greater focus on the literature in terms of continuous systems than their discrete-time counterpart. For the works that consider the system as discrete-time [18], linear time-varying systems [19], stochastic - [20], markovian-jump stochastic systems [21] and more general algebraic and difference equations [22] have been considered. However, in all the previous cases where the system was considered as discrete-time, the inputs were assumed to be measured or known.

As such, this work extends the concepts of observability to discrete-time systems with unmeasured inputs. It should be noted that the observability properties obtained would be affected by the parameters used to describe the inputs. However, for ease of presentation, the work assumes that the inputs to the system can be perfectly represented by using a zero-order-hold over some sampling frequency. It is further assumed that the outputs are logged at the same time steps as the inputs. These assumptions are made as they are commonly used in input estimation algorithms.

While the observability algorithm is presented initially for systems that are discrete-time, later sections discuss the case of continuous-time systems that have been approximated by some discretization. Time integration methods of different order are considered, where the integration time step is allowed to be an integer multiple of the sampling time step. In the cases where the system is approximated, technically the observability investigated is that of the examined model used for estimation rather than the underlying dynamic system that generated the data, which as stated in as defined in [9] should be referred to as model observability. However, since identification algorithms always use an estimation model without knowing the ‘true’ model that generated the data, it is the properties of that model that are always of interest. Thus from here-after only the term observability will be used to refer to the properties of any of the models studied. Nonetheless, the observability of the perfectly discretized system can be obtained for a sufficiently high integration order. The rows of the observability matrix are linked to different time steps at which measurements are obtained. The observability properties refer to the initial conditions of the states, the parameters, and the piecewise constants that represent the input at different time instances following the ZOH discretization. This allows for defining a new set of properties, that describe at which measurement step each of the previous quantities becomes observable for the first time. The new concept of a lag for estimating inputs is defined and its importance to input estimation algorithms is explained.

Following the description of the new algorithm in Section 2, a series of examples demonstrate the new concept of the observability of discrete-time systems in Section 3. The effects of discretization and sensor setup on the observability properties of the system are highlighted. The observability properties of the discretized systems are compared to that of the corresponding continuous systems, while links are made to commonly employed requirements for certain input estimation algorithms, e.g., it is shown that the commonly assumed requirement for collocation of the unmeasured force and excitation is a result of the chosen discretization of the system.

2. Methodology

This section will present the new concept of observability of discrete-time systems involving unmeasured inputs. This concept will be more readily applicable to estimation models used in system identification algorithms. To draw comparisons to the corresponding concept of the observability of continuous-time systems the structure of the proofs will be presented in a similar manner to [13,14].

Consider a system driven by n_u measured inputs $u^{(i)}$, n_p unmeasured inputs $p^{(i)}$, and n_y measured outputs $y^{(i)}$, as schematically represented in Fig. 1. Without loss of generality, the dynamic behaviour of this system can be mathematically described by the following nonlinear continuous-time state-space representation:

$$\dot{\mathbf{x}} = \boldsymbol{\phi}(\mathbf{x}, \theta, \mathbf{u}, \mathbf{p})\mathbf{y} = \boldsymbol{h}(\mathbf{x}, \theta, \mathbf{u}, \mathbf{p}) \tag{1}$$

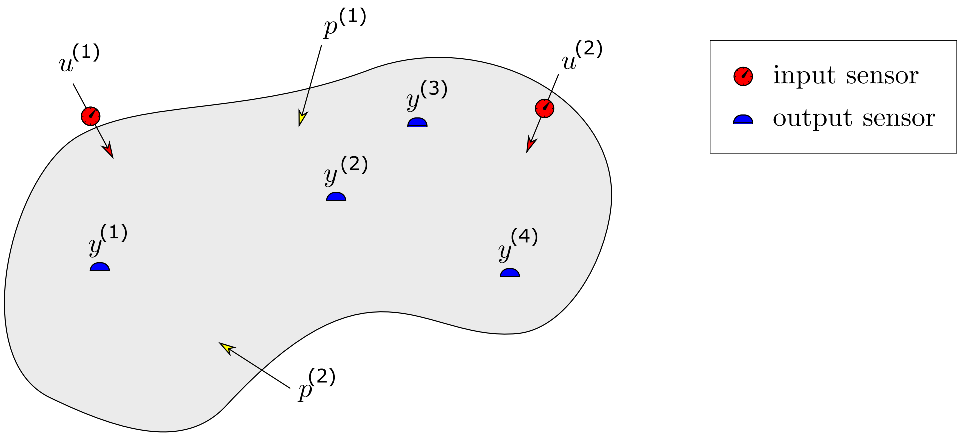


Fig. 1. Schematic representation of a dynamic system with measured inputs $u^{(i)}$, unmeasured inputs $p^{(i)}$, and measured outputs $y^{(i)}$.

where $\mathbf{x}$ is the state vector of size n_{s_t}, $\mathbf{u}$ is a vector of n_u measured inputs, $\mathbf{p}$ is a vector of n_p unmeasured inputs, $\boldsymbol{\theta}$ is the vector of n_t time-invariant parameters, $\mathbf{y}$ is the output vector of size n_y, and $\boldsymbol{\phi}$ and $\boldsymbol{h}$ are nonlinear vector functions of dimension n_{s_t} and n_y, respectively. Additionally, the size of the state and parameter augmented vector $[\mathbf{x}, \boldsymbol{\theta}]$ will be referred to as n_s, i.e., $n_s = n_{s_t} + n_t$. At this point, it is worth noting that the assumption of the parameters being time invariant in Eq. (1) is not restrictive, as a parameter which would be time-variant, depending on whether its evolutionary equation is known, would either be treated as dynamic state or as an input, respectively.

The system of Fig. 1 can also be represented by a discrete-time representation which would describe the states $\mathbf{x}_k$ and outputs $\mathbf{y}_k$ of the system evaluated at the sampling time steps $t_k = k\,\Delta t$, $k = 0, 1, 2, \ldots$, where Δt is the sampling time step. In this work, the measured and unmeasured inputs will be sampled under the assumption of zero-order-hold. The equations for such a discrete-time system representation can be presented as:

$$\mathbf{x}_{k+1} = \boldsymbol{\Phi}(\mathbf{x}_k, \boldsymbol{\theta}, \mathbf{U}_k, \mathbf{P}_k)\mathbf{y}_{k+1} = \boldsymbol{h}(\mathbf{x}_{k+1}, \boldsymbol{\theta}, \mathbf{U}_{k+1}, \mathbf{P}_{k+1}) \tag{2}$$

where $\mathbf{U}_k$ and $\mathbf{P}_k$ would correspond to $\mathbf{u}$ and $\mathbf{p}$, respectively evaluated at the time instance t_k. The function $\boldsymbol{\Phi}$ represents the flow function. For reasons of brevity, the flow function from t_k to $t_k + 1$, $\boldsymbol{\Phi}(\mathbf{x}_k, \boldsymbol{\theta}, \mathbf{U}_k, \mathbf{P}_k)$, will be denoted as: $\boldsymbol{\Phi}_k(\mathbf{x}_k)$. It should be noted that $\boldsymbol{\Phi}_k(\mathbf{x}_k)$ depends not only on $\mathbf{x}_k$ but also on $\boldsymbol{\theta}$, $\mathbf{U}_k$ and $\mathbf{P}_k$.

For a given realization of the initial conditions $\mathbf{x}_0$, parameters $\boldsymbol{\theta}$, piecewise constant measured inputs $\mathbf{U}_{k+1}$, and unmeasured inputs $\mathbf{P}_{k+1}$, the state $\mathbf{x}$ at time steps $1, \ldots, k$ can be evaluated through successive application of the flow function $\boldsymbol{\Phi}$:

$$\begin{aligned}
\mathbf{x}_1 &= \boldsymbol{\Phi}(\mathbf{x}_0, \theta, \mathbf{U}_0, \mathbf{P}_0) = \boldsymbol{\Phi}_1(\mathbf{x}_0) \\
\mathbf{x}_2 &= \boldsymbol{\Phi}(\mathbf{x}_1, \theta, \mathbf{U}_1, \mathbf{P}_1) = \boldsymbol{\Phi}_2 \circ\ \boldsymbol{\Phi}_1(\mathbf{x}_0) \\
&\vdots \\
\mathbf{x}_k &= \boldsymbol{\Phi}_k \circ \ldots \circ \boldsymbol{\Phi}_1(\mathbf{x}_0)
\end{aligned} \tag{3}$$

It can be seen that according to Eq. (3), $\mathbf{x}_k$ can be expressed as function of $\mathbf{x}_0$, $\boldsymbol{\theta}$, $\mathbf{U}_0, \ldots, \mathbf{U}_{k-1}$ and $\mathbf{P}_0, \ldots, \mathbf{P}_{k-1}$. However, due to the causal nature of the inputs and their representation through zero-order-hold, $\mathbf{x}_k$ is independent of $\mathbf{U}_k, \ldots, \mathbf{U}_\infty$ and $\mathbf{P}_k, \ldots, \mathbf{P}_\infty$.

Similarly, the outputs y_k at time steps $0, \ldots, k$ are evaluated as:

$$\begin{aligned}
\mathbf{y}_0 &= \boldsymbol{h}(\mathbf{x}_0, \boldsymbol{\theta}, \mathbf{U}_0, \mathbf{P}_0) \\
\mathbf{y}_1 &= \boldsymbol{h}(\mathbf{x}_1, \boldsymbol{\theta}, \mathbf{U}_1, \mathbf{P}_k) = \boldsymbol{h}(\boldsymbol{\Phi}_1(x_0), \boldsymbol{\theta}, \mathbf{U}_1, \mathbf{P}_1) \\
&\vdots \\
\mathbf{y}_k &= \boldsymbol{h}(\mathbf{x}_k, \boldsymbol{\theta}, \mathbf{U}_k, \mathbf{P}_k) = \boldsymbol{h}(\boldsymbol{\Phi}_k \circ \ldots \circ \boldsymbol{\Phi}_1(\mathbf{x}_0), \boldsymbol{\theta}, \mathbf{U}_k, \mathbf{P}_k)
\end{aligned} \tag{4}$$

As can be seen from Eq. (4), $\mathbf{y}_k$ can be expressed as function of $\mathbf{x}_0$, $\boldsymbol{\theta}$, $\mathbf{U}_0, \ldots, \mathbf{U}_k$ and $\mathbf{P}_0, \ldots, \mathbf{P}_k$. It is however independent of all future inputs $\mathbf{U}_j$ and $\mathbf{P}_j$ with $j > k$.

This motivates defining the augmented vector $\mathbf{z}^k = [\mathbf{x}_0, \boldsymbol{\theta}, \mathbf{P}_0, \ldots, \mathbf{P}_k]$. It should be noted that $\mathbf{z}^k$ comprises the initial conditions of the dynamic states $\mathbf{x}_0$ and all values of the unmeasured inputs from t_0 to t_k. Since the parameters $\boldsymbol{\theta}$ are assumed time-invariant, their initial values at t_0 match their values at any other time instance.

Lemma 2.1. *Two state vectors $\mathbf{z}^{k^{\mathrm{a}}}$ and $\mathbf{z}^{k^{\mathrm{b}}}$ are indistinguishable if they lead to the same output $\mathbf{y}$ at any time $t = \bar{t}_j$ for the measured piecewise constant inputs $\mathbf{u}$, i.e., if:*

$$\begin{aligned}
\mathbf{y}_j^{\mathrm{a}} &= \mathbf{y}_j^{\mathrm{b}} \\
\boldsymbol{h}(\boldsymbol{\Phi}_j^{\mathrm{a}} \circ \ldots \circ \boldsymbol{\Phi}_1^{\mathrm{a}}(\mathbf{x}_0{}^{\mathrm{a}}), \theta^{\mathrm{a}}, \mathbf{U}_j, \mathbf{P}_j^{\mathrm{a}}) &= \boldsymbol{h}(\boldsymbol{\Phi}_j^{\mathrm{b}} \circ \ldots \circ \boldsymbol{\Phi}_1^{\mathrm{b}}(\mathbf{x}_0{}^{\mathrm{b}}), \theta^{\mathrm{b}}, \mathbf{U}_j, \mathbf{P}_j^{\mathrm{b}})
\end{aligned} \tag{5}$$

Definition 2.1. The states $\mathbf{z}^k$ are classified as k-row observable if their value can be separated locally from its neighbours based on the output $\mathbf{y}$ at $k+1$ consecutive times $t = \bar{t}_0, \bar{t}_1, \dots, \bar{t}_k$, meanwhile considering every possible input $\mathbf{u}$ over the considered time interval.

The following column vector $\boldsymbol{\Omega}_k$ can then be defined:

$$\boldsymbol{\Omega}_k = \begin{bmatrix} \mathbf{y}_0 \\ \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_k \end{bmatrix} \tag{6}$$

and its derivative $d\boldsymbol{\Omega}_k$ with respect to $\mathbf{z}^k$:

$$d\boldsymbol{\Omega}_k = \nabla_{\mathbf{z}^k}(\boldsymbol{\Omega}_k) = \nabla_{\mathbf{x}_0,\theta,\mathbf{P}_0,\dots,\mathbf{P}_k}(\boldsymbol{\Omega}_k) = \begin{bmatrix} \nabla_{\mathbf{x}_0}(\boldsymbol{\Omega}_k)\nabla_{\theta}(\boldsymbol{\Omega}_k) & \nabla_{\mathbf{P}_0}(\boldsymbol{\Omega}_k) & \cdots & \nabla_{\mathbf{P}_k}(\boldsymbol{\Omega}_k) \end{bmatrix} \tag{7}$$

As in [13,14] the inverse function theorem [23] ensures that if $\mathrm{rank}(\mathrm{d}\boldsymbol{\Omega}_k) = n_\mathrm{s} + (k+1)n_\mathrm{p}$ the system is k−row observable with respect to all the dynamic states x_0, $\boldsymbol{\theta}$ and each of the piecewise-constant inputs that have appeared up to the time instance t_k: $\mathbf{P}_0, \dots, \mathbf{P}_k$.

If the k−row observability matrix for a chosen $k = k_0$ is not a full-rank matrix, i.e. $\mathrm{rank}(\mathrm{d}\boldsymbol{\Omega}_k) < n_\mathrm{s} + (k+1)n_\mathrm{p}$, it is then of interest to investigate the observability property of each component of $\mathbf{z}^k$. This can be achieved by removing the ith column of $d\boldsymbol{\Omega}_k$. If the rank of the occurring matrix is smaller than the rank of $d\boldsymbol{\Omega}_k$, then the ith element (dynamic state/parameter/unmeasured input) of $\mathbf{z}^k$ is observable; otherwise it is k−row unobservable as defined in [13].

It should be noted that a k_0-row unobservable state might become k−row observable when using a $k > k_0$, i.e., after adding more measurements at later time steps. In other words, k−row observability guarantees observability, but k−row unobservability does not guarantee unobservability. This is a property of many continuous observability algorithms under the presence of unmeasured inputs, except the algorithm suggested in [16] which allows reaching a stopping criterion where a property can be guaranteed to remain unobservable. However, for discrete-time systems, the information that a parameter has not become observable by a specific measurement step is directly useful as will be discussed later in the paper.

Taking into account the aforementioned considerations, the following algorithm can be applied to verify the observability of the discrete-time system in the presence of unmeasured inputs.

Algorithm 2.1.
Starting point: $k = 0$, $\mathbf{z}^0 = \begin{bmatrix}\mathbf{x}_0^\mathrm{T} & \boldsymbol{\theta}^\mathrm{T} & \mathbf{P}_0{}^\mathrm{T}\end{bmatrix}^\mathrm{T}$.

1. Compose $\Delta\boldsymbol{\Omega}_0 = \begin{bmatrix}\mathbf{y}_0\end{bmatrix}$.
2. Calculate $\mathrm{d}^0\boldsymbol{\Omega}_0 = \frac{\partial\boldsymbol{\Omega}_0}{\partial\mathbf{z}^0}$
3. $k = k + 1$.
4. Calculate $\Delta\boldsymbol{\Omega}_k = \begin{bmatrix}\mathbf{y}_k\end{bmatrix}^\mathrm{T}$, from equation (4)
5. Compose $\mathbf{z}^k = \begin{bmatrix}\mathbf{z}^{k-1\mathrm{T}} & \mathbf{P}_k^T\end{bmatrix}^\mathrm{T}$.
6. Calculate $\mathrm{d}^k\Delta\boldsymbol{\Omega}_k = \frac{\partial\Delta\boldsymbol{\Omega}_k}{\partial\mathbf{z}^k}$.
7. Calculate $\mathrm{d}^k\boldsymbol{\Omega}_k = [\mathrm{d}^{k-1}\boldsymbol{\Omega}_{k-1} \quad \mathbf{0}] \cup \mathrm{d}^k\Delta\boldsymbol{\Omega}_k$.
8. Optional step: eliminate dependent rows from $\mathrm{d}^k\boldsymbol{\Omega}_k$.
9. If $\mathrm{rank}(\mathrm{d}^k\boldsymbol{\Omega}_k) = n_\mathrm{s} + (k+1)n_\mathrm{p}$, end.
10. Investigate partial observability for the m-th state of $\mathbf{z}^k$.
    a. Starting point: $m = 1$.
    b. If the m^{th} state is $(k-1)$-row observable, it is also k-row observable. Go to step f.
    c. Compose $\mathrm{d}^k\boldsymbol{\Omega}_k^m$ by removing the m^{th} column from $\mathrm{d}^k\boldsymbol{\Omega}_k$.
    d. The m^{th} state is k-row observable if and only if $\mathrm{rank}(\mathrm{d}^k\boldsymbol{\Omega}_k^m) < \mathrm{rank}(\mathrm{d}^k\boldsymbol{\Omega}_k)$.
    e. If $m = n_\mathrm{s} + (k-1)n_\mathrm{p}$, end.
    f. m=m+1 and go to step b.
11. If the elements of $\mathbf{z}^k$ are k-row observable, end.
12. Go to step 3.

It should be noted that although step 10 in the above algorithm allows for investigating the k− row observability of any element in $\mathbf{z}^k$, this observability takes into account the effect of the other elements in $\mathbf{z}^k$. This is not equivalent to studying the observability of that element in $\mathbf{z}^k$ by assuming other elements in $\mathbf{z}^k$ be known. For example, when a state is found observable in an observability analysis where all inputs are known/measured, this does not guarantee that the state is observable in the more general case investigated by the above algorithm where the effect of $\mathbf{P}$ is taken into account.

As was previously stated in [17] for continuous-systems, the number of unobservable elements of the augmented state vector is generally greater than the number of groups of Lie-symmetries those unobservable quantities participate in. This additional step of detecting the Lie-symmetries of a discrete-time system is a future direction of research, but is beyond the scope of this work.

Lemma 2.2. *If $p_{k_1}^{(i)}$ is $k-$row observable, then $p_{k_2}^{(i)}$ for $k_2 \geq k_1$ will also be $k-$row observable.*

Proof. Let Σ_0 be the dynamic system defined by Eq. (2), under the influence of the inputs $\mathbf{P}_j$, $\mathbf{U}_j$, with initial conditions at t_0, x_0, and the k_1+1 measurements obtained starting from t_0: $[\mathbf{y}_0, \dots, \mathbf{y}_{k_1}]$. The augmented state for that system at step k_1 for Σ_0 is defined as: $\mathbf{z}^{k_1} = [\mathbf{x}_0, \theta, \mathbf{P}_0, \dots, \mathbf{P}_{k_1}]$ and the $d\boldsymbol{\Omega}_{k_1}$ is defined from Eq. (7). For the same system, measurements are obtained until the future step k_2, with $k_2 > k_1$, $[\mathbf{y}_0, \dots, \mathbf{y}_{k_2}]$, under the influence of the inputs $\mathbf{P}_j$, $\mathbf{U}_j$, $j = 0, \dots, k_2$. The system $\Sigma_{k_2-k_1}$ is obtained from Σ by defining the starting time at time step $t_{k_2-k_1}$ and the corresponding initial conditions of the states, $\mathbf{x}'_0 = \mathbf{x}_{k_2-k_1}$. For $\Sigma_{k_2-k_1}$, the relevant measurements are $\boldsymbol{\Omega}'_{k_1} = [\mathbf{y}_{k_2-k_1}, \dots, \mathbf{y}_{k_2}]$ and have been generated under the influence of x'_0, θ and the inputs $\mathbf{P}'_j = \mathbf{P}_{j+k_2-k_1}$, $\mathbf{U}'_j = \mathbf{U}_{j+k_2-k_1}$, $j = 0, \dots, k_2$. Defining the augmented state vector, $\mathbf{z}'^{k_1} = [\mathbf{x}'_0, \theta, \mathbf{P}'_0, \dots, \mathbf{P}'_{k_1}]$, for $\Sigma_{k_2-k_1}$ results according to Eq. (7) in a matrix $d\boldsymbol{\Omega}'_{k_1}$. It is then clear that the only differences between $d\boldsymbol{\Omega}'_{k_1}$ and $d\boldsymbol{\Omega}_{k_1}$ are the potential realizations of z'_0 and $\mathbf{U}'_j$, versus z_0 and $\mathbf{U}_j$. However, we are interested in the general rank properties of $d\boldsymbol{\Omega}'_{k_1}$ and $d\boldsymbol{\Omega}_{k_1}$ which in the general case of not choosing realizations for which the rank property is reduced would be independent of the numerical values of those realizations. As such, the two matrices $d\boldsymbol{\Omega}'_{k_1}$ and $d\boldsymbol{\Omega}_{k_1}$ have the same rank properties. By construction, the jth state of z^{k_1} and z'^{k_1} corresponds to the jth column of $d\boldsymbol{\Omega}_{k_1}$ and $d\boldsymbol{\Omega}'_{k_1}$ respectively and have the same observability properties. □

The previous result can also be complimented, to the relatively obvious result that any element of z_0 that becomes $k-$ row observable at some measurement time step, will remain observable for all future time steps.

Definition 2.2. If the jth element of $\mathbf{x}_0$, x_{0_j}, the parameter θ_j, or the unmeasured piecewise constant input $p_j^{(i)}$ became $k-$row observable for the first time at time step t_{k_1}, i.e, after collecting k_1+1 measurement steps, the following quantities are defined: $N_{x_{0_j}} = k_1+1$, $N_{\theta_j} = k_1+1$ and $N_{p_j^{(i)}} = k_1+1$, respectively.

It should be noted that in the previous definition the subscript j could refer to jth element of the state vector $\mathbf{x}_0$, or the vector of parameters θ, or the jth piecewise-constant of input $p^{(i)}$. It is reminded that for the latter element the subscript j also has the meaning of time, i.e., the corresponding piecewise constant defined at time step t_j. However for the first two vectors the ordering of their elements is not associated to time and is instead a non-unique to a physical-system choice made when defining those vectors.

Corollary 2.1. *If $N_{x_j} = k$, $N_{\theta_j} = k$ or $N_{p_j^{(i)}} = k$, the corresponding state/parameter/piecewise constant input can be properly estimated only after at least gathering the measurements* $[\mathbf{y}_0, \dots, \mathbf{y}_{k-1}]$.

Definition 2.3. If the unmeasured piecewise constant input $p_j^{(i)}$ becomes observable at some step, its lag $L_{p_j^{(i)}}$, is defined as $L_{p_j^{(i)}} = N_{p_j^{(i)}} - j - 1$.

Based on the above Definition 2.3, the lag for $p_j^{(i)}$ is the minimum number of measurement equations occurring at time steps after its appearance at the $j+1^{th}$ time step, before $p_j^{(i)}$ can be properly estimated. It is common practice in several online input estimation filters to involve each value of $p_j^{(i)}$ in a single time-update equation resulting in it affecting explicitly at most the measurement equations of time-steps t_j and t_{j+1}. When $L_{p_j^{(i)}} > 1$, such algorithms do not involve the piecewise constants in a sufficient number of measurement steps. Overall, larger values of $L_{p_j^{(i)}}$ poses greater challenges to input estimation algorithms.

Using Definition 2.3 and Lemma 2.2 the following corollary can be obtained:

Corollary 2.2. *If $j_1 < j_2$, then $L_{p_{j_1}^{(i)}} \geq L_{p_{j_2}^{(i)}}$.*

And following Corollary 2.2, if the unmeasured piecewise constant input $p_j^{(i)}$ becomes observable at some step, there will be a measurement step j_m at which $L_{p_j^{(i)}}$ attains its minimum value, $L_{p^{(i)}}$, $L_{p^{(i)}} = L_{p_{j_m}^{(i)}}$. Then following Corollary 2.2, for any measurement step $j \geq j_m$ $L_{p_j^{(i)}} = L_{p^{(i)}}$.

In the case where all the unmeasured inputs of the system are observable, the maximum value of $L_{p^{(i)}}$ of the unmeasured inputs $p^{(i)}$ of the system can be defined as, L:

Definition 2.4. $L = \max_i(L_{p^{(i)}})$

Finally, a situation that occurs often is this where the states and parameters of the system become all observable. If this occurs at t_{k-1}, i.e., after k time steps, this will be defined by $N_s = k$. This quantity is also of interest as it indicates that to properly estimate at least all of the $\mathbf{x}_0$ and θ, a common goal in System Identification, one needs to obtain at least the measurements $[\mathbf{y}_0, \dots, \mathbf{y}_{N_s}]$.

2.1. Discretization in time of a continuous dynamic system

Eq. (2) provides a direct description of the system of Fig. 1 in discrete-time. For some systems that can be obtained directly, e.g., if the system is linear in terms of the dynamic states $\mathbf{x}$, then the ZOH rule would allow for an exact discretization. But it is often the case that a perfect description of the system can only be obtained in continuous-time according to Eq. (1), and then the Discrete-time description of the system is obtained following a discretization rule.

The simplest integration rule, which is nonetheless often used in models used for System Identification, is the Euler integration rule which can be expressed as:

$$\boldsymbol{\Phi}(\mathbf{x}_k, \theta, \mathbf{U}_k, \mathbf{P}_k) = \mathbf{x}_k + \boldsymbol{\phi}(\mathbf{x}_k, \theta, \mathbf{U}, \mathbf{P})\, \Delta t \tag{8}$$

More general integration rules can be used, such as various explicit and implicit time integration methods [24]. Several of these methods involve splitting the sampling time step Δt into smaller integration time steps. Without loss of generality, and to facilitate comparisons of the effect of the integration step, this work focuses on methods that break the sampling time step Δt into N integration time steps $dI = \Delta t/N$, and define the flow for the $N-th$ order integration rule as:

$$\begin{aligned} \mathbf{x}_{k^{l+1}} &= \mathbf{x}_{k^l} + \boldsymbol{\phi}(\mathbf{x}_{k^l}, \theta, \mathbf{U}_k, \mathbf{P}_k)\, \Delta t/N, l = 0, \dots, N-1 \\ \boldsymbol{\Phi}(\mathbf{x}_k, \theta, \mathbf{U}_k, \mathbf{P}_k) &= \mathbf{x}_{k^N} \end{aligned} \tag{9}$$

where $\mathbf{x}_{k^0} = \mathbf{x}_k$ and $\mathbf{x}_{k+1} = \mathbf{x}_{k^N}$. When $N = 1$, the Euler integration scheme shown in Eq. (8) is obtained. The flow, $\boldsymbol{\Phi}(\mathbf{x}_k, \theta, \mathbf{U}_k, \mathbf{P}_k)$ defined in Eq. (9) is an approximation of the exact discretization of the continuous system in Eq. (1) which would technically be obtained from this rule if $N \to \infty$. It should be noted that the sampling frequency of the inputs and outputs is independent of N.

If one considers that the system that generates the data is the continuous system (1) together with a sampling process that obtains the data at discrete points, then the observability of interest would correspond to the exact integration for $N \to \infty$. Then the approximations described in Eq. (9) would correspond to different models and technically the use of an observability algorithm on such models would result in what is described in [9] as model-observability. This is to indicate that by changing the $N-th$ order discretization scheme, one only changes the observability of the discrete-time model that is being attempted to be identified, and not the observability of the physical system that generated the data. For the sake of brevity, in this work, we will also refer to model-observability as observability.

It then becomes clear that the (model) observability of the system depends on the discretization scheme used. Similarly, N_s and $L_{p^{(i)}}$, if they exist, will also depend on N. It will also be shown through the examples, that there exist finite values of N, N_f, where the further increase of N does not result in any change of the observability properties, including the values of N_s and $L_{p^{(i)}}$ (if they exist). Then the observability property of the exact discretization of the system is the same as those for $N = N_f$.

It should finally be noticed that the continuous observability properties of a system would not necessarily match the discrete observability properties of the discrete system, as the latter can describe situations where the sampling step can potentially be small but not infinitesimally small, as is the case for the former.

2.2. Recursion

Algorithm 2.1 has the computational advantage over its continuous counterpart, as e.g., presented in [13], that it does not require the calculation of higher order derivatives. Nonetheless, even the recurring calculation of the flow function $\boldsymbol{\Phi}$, which is required in Eq. (4) successively results [25] in swelling of the symbolic expressions, and renders the direct calculation of the corresponding terms of $\mathrm{d}^k \Delta\boldsymbol{\Omega}_k = \frac{\partial \Delta\boldsymbol{\Omega}_k}{\partial \mathbf{z}^k}$ computationally expensive for modest values of k. This dictates the recursive calculation of those derivatives, which is shown in the following algorithm:

Algorithm 2.2.

Starting point: $k = 0$, Calculate $[\nabla_{\mathbf{x}_0}\mathbf{y}_0, \nabla_\theta \mathbf{y}_0, \frac{\partial \mathbf{y}_0}{\partial p_0^1}, \cdots, \frac{\partial \mathbf{y}_0}{\partial p_0^{n_p}}, \mathbf{0}]$.

1. $\mathbf{x}_{k+1} = \boldsymbol{\Phi}(\mathbf{x}_k, \theta, \mathbf{U}_k, \mathbf{P}_k)$
2. $\nabla_{\mathbf{x}_0}\mathbf{x}_{k+1} = \nabla_{\mathbf{x}_k}\mathbf{x}_{k+1}\, \nabla_{\mathbf{x}_0}\mathbf{x}_k = \nabla_{\mathbf{x}}\boldsymbol{\Phi}\Big|_{\mathbf{x}=\mathbf{x}_k} \nabla_{\mathbf{x}_0}\mathbf{x}_k$
3. $\nabla_\theta \mathbf{x}_{k+1} = \nabla_{\mathbf{x}}\boldsymbol{\Phi}\Big|_{\mathbf{x}=\mathbf{x}_k} \nabla_\theta \mathbf{x}_k + \nabla_\theta \boldsymbol{\Phi}\Big|_{\mathbf{x}=\mathbf{x}_k}$
4. $\frac{\partial x_{k+1}}{\partial p_j^n} = \nabla_{\mathbf{x}}\boldsymbol{\Phi}\Big|_{\mathbf{x}=\mathbf{x}_k} \frac{\partial x_k}{\partial p_j^n} + \frac{\partial \boldsymbol{\Phi}}{\partial p_j^n}\Big|_{\mathbf{x}=\mathbf{x}_k}$
5. $\nabla_{\mathbf{x}_0}\mathbf{y}_{k+1} = \nabla_{\mathbf{x}}\boldsymbol{h}\Big|_{\mathbf{x}=\mathbf{x}_{k+1}} \nabla_{\mathbf{x}_0}\mathbf{x}_{k+1}$
6. $\nabla_\theta \mathbf{y}_{k+1} = \nabla_{\mathbf{x}}\boldsymbol{h}\Big|_{\mathbf{x}=\mathbf{x}_{k+1}} \nabla_\theta \mathbf{x}_{k+1} + \nabla_\theta \boldsymbol{h}\Big|_{\mathbf{x}=\mathbf{x}_{k+1}}$
7. $\frac{\partial y_{k+1}}{\partial p_j^n} = \nabla_{\mathbf{x}}\boldsymbol{h}\Big|_{\mathbf{x}=\mathbf{x}_k+1} \frac{\partial x_{k+1}}{\partial p_j^n} + \frac{\partial \boldsymbol{h}}{\partial p_j^n}\Big|_{\mathbf{x}=\mathbf{x}_{k+1}}$
8. $d\boldsymbol{\Omega}_{k+1} = d\boldsymbol{\Omega}_k \cup [\nabla_{\mathbf{x}_0}\mathbf{y}_{k+1}, \nabla_\theta \mathbf{y}_{k+1}, \frac{\partial \mathbf{y}_{k+1}}{\partial p_0^1}, \cdots, \frac{\partial \mathbf{y}_{k+1}}{\partial p_{k+1}^{n_p}}, \mathbf{0}]$
9. Set $k = k + 1$ and go to step 1.

The recursive steps in Algorithm 2.2 require only the calculation of derivatives $\nabla_{\mathbf{x}}\boldsymbol{\Phi}$, $\nabla_{\theta}\boldsymbol{\Phi}$, $\frac{\partial \boldsymbol{\Phi}}{\partial p^n}$, $\nabla_{\mathbf{x}}\boldsymbol{h}$, $\nabla_{\theta}\boldsymbol{h}$ and $\frac{\partial h}{\partial p^n}$. The derivatives can be calculated symbolically once and then are evaluated at the corresponding values of $\mathbf{x} = \mathbf{x}_j$, θ, $\mathbf{U} = \mathbf{U}_k$ and $\mathbf{P} = \mathbf{P}_k$. Those derivatives are often simple to evaluate, but especially the derivatives of $\boldsymbol{\Phi}$ can become computationally expensive especially when the integration order N grows.

To alleviate this computational cost, a second layer of recursion is defined which allows for the recursive computation of those terms. This is described in the following algorithm:

Algorithm 2.3.

Starting point: $k = 0$,

1. $\mathbf{x}_{k^{l+1}} = \boldsymbol{\Phi}_N(\mathbf{x}_{k^l}, \theta, \mathbf{U}_k, \mathbf{P}_k)$
2. $\nabla_{\mathbf{x}_k}\mathbf{x}_{k^{l+1}} = \nabla_{\mathbf{x}_{k^l}}\mathbf{x}_k^{l+1}\,\nabla_{\mathbf{x}_k}\mathbf{x}_k^{l} = \nabla_{\mathbf{x}}\boldsymbol{\Phi}_N\Big|_{\mathbf{x}=\mathbf{x}_{k^l}}\nabla_{\mathbf{x}_k}\mathbf{x}_{k^l}$
3. $\nabla_{\theta}\mathbf{x}_{k^{l+1}} = \nabla_{\mathbf{x}}\boldsymbol{\Phi}_N\Big|_{\mathbf{x}=\mathbf{x}_{k^l}}\nabla_{\theta}\mathbf{x}_{k^l} + \nabla_{\theta}\boldsymbol{\Phi}_N\Big|_{\mathbf{x}=\mathbf{x}_{k^l}}$
4. $\frac{\partial x_{k^{l+1}}}{\partial p_j^n} = \nabla_{\mathbf{x}}\boldsymbol{\Phi}_N\Big|_{\mathbf{x}=\mathbf{x}_{k^l}}\frac{\partial x_{k^l}}{\partial p_j^n} + \frac{\partial \boldsymbol{\Phi}_N}{\partial p_j^n}\Big|_{\mathbf{x}=\mathbf{x}_{k^l}}$
5. Set $k = k + 1$ and go to step 1.

Through the use of recursion, the two Algorithms 2.3 and 2.2 allow for calculating the entries of $d\boldsymbol{\Omega}_{k+1}$ by propagating numerical values of realizations of $\mathbf{x}_0$, θ, $\mathbf{U}_j$ and $\mathbf{P}_j$. This motivates the following subsection regarding the calculation of a rank of a numerical matrix whose entries have been calculated using finite precision arithmetics.

2.3. Numerical calculation of rank of $d\boldsymbol{\Omega}$

The calculation of $d\boldsymbol{\Omega}_k$ through the use of Algorithm 2.2 and potentially 2.3 involve a series of mathematical operations described by the various levels of recursion. Such operations are exact when performed with symbolic numbers, i.e., with exact and infinite precision arithmetic. However, doing so results in a significant increase in the computational requirements. As the number of operations increases a greater number of digits is generally required for the exact representation of numbers. This rapidly results in substantial requirements of physical memory (RAM) which often exceed what is typically available in a contemporary workstation. Overall, this process is not computationally viable for systems of modest size.

In this work, finite precision arithmetic is used for the calculation of the rank properties of $d\boldsymbol{\Omega}_k$ and the relevant sub-matrices used for observability. In doing so, the following lines will present a procedure for correctly estimating the rank properties despite the errors introduced by finite precision arithmetic operations.

Without loss of generality a numerical realization of $\mathbf{x}_0, \theta, \mathbf{P}_j$ and $\cdots, \mathbf{U}_j$, $j = 0, \ldots, \infty$ is chosen so that the chosen numbers can be perfectly represented with a precision of up to nd digits. This choice should not influence the rank as it is statistically unlikely that such a random choice of that realization would happen to match one of the singular realizations of $d\boldsymbol{\Omega}_k$. At the same time any parameter appearing in $\boldsymbol{\phi}$ which is not to be identified is also specified with a precision of up to nd_1 digits. In the common case, where such a parameter corresponds to some geometric angle, maximum precision is required and hence nd digits would be used. The calculation of $d\boldsymbol{\Omega}_k$ at step k will involve a series of operations that will also be done with the precision of nd_1 digits. As a consequence, the following equation describes the numerically imprecise matrix, $d\boldsymbol{\Omega}'_k$, obtained through this process:

$$d\boldsymbol{\Omega}_{k_{nd_1}} = d\boldsymbol{\Omega}_k + \epsilon_{d\boldsymbol{\Omega}_k} \tag{10}$$

where $\epsilon_{d\boldsymbol{\Omega}_k}$ is the error matrix $d\boldsymbol{\Omega}_{k_{nd_1}} - d\boldsymbol{\Omega}_k$. It should be noticed that $d\boldsymbol{\Omega}'_k$ is formed after multiple operations of nd_1 precision and as such there is no guarantee of the order of the elements $\epsilon_{d\boldsymbol{\Omega}_k}$.

As an example, the following Figs. 2(a) show the logarithm of the difference from two $d\boldsymbol{\Omega}'_k$ matrices corresponding to the same problem but calculated with nd_1 corresponding to 90 and 100 digits respectively. The horizontal axis of the Figure corresponds to the columns and the vertical axis corresponds to the rows of the corresponding matrices. While the differences are zero for the operations of the first few rows, they become increasingly larger as more rows are added. However, from Fig. 2(b) it can also be seen that the relative differences are affected in a different manner for different terms.

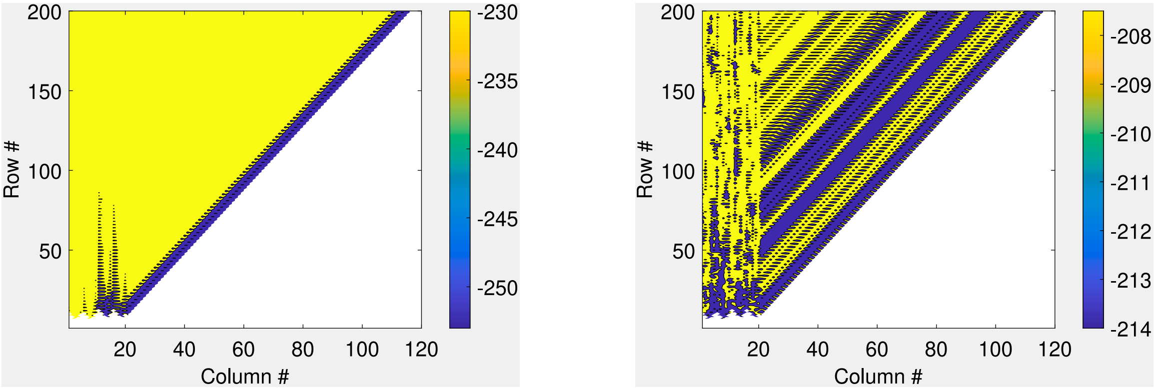
To find the rank of the occurring matrix $d\boldsymbol{\Omega}'$, a numerical method can be used. The most commonly used in the literature is that of estimating the singular values of $d\boldsymbol{\Omega}'_k$:

$$\mathbf{SV}_{nd_2}(d\boldsymbol{\Omega}_k) = \mathbf{SV}(d\boldsymbol{\Omega}_k) + \epsilon_{SV(d\boldsymbol{\Omega}_k)} \tag{11}$$

where $\mathbf{SV}_{nd}$ is the operation of estimating the singular values of a matrix using operations of up to nd_2 digits precisions and $\mathbf{SV}$ is the infinite precision operation of estimating the singular values of a matrix. This last source of error is hard to estimate as it would require a means of perfectly estimating the singular values of the exact matrix $d\boldsymbol{\Omega}_k$ and so it is more practical to combine the two sources of error in the following equation:

$$\mathbf{SV}_{nd_2}(d\boldsymbol{\Omega}_{k_{nd_1}}) = \mathbf{SV}(d\boldsymbol{\Omega}_k) + \epsilon_{nd_1,nd_2} \tag{12}$$

An example of how the Singular Values of a given matrix change as the number of digits nd_2 changes is shown in the Figure below.



(a) Difference   (b) Relative difference

Fig. 2. The natural logarithm of the difference (a) and the relative difference (b) of the terms of the $d\boldsymbol{\Omega}_k$ matrix calculated with two different $nd_1 = 90$ and 100 digits.

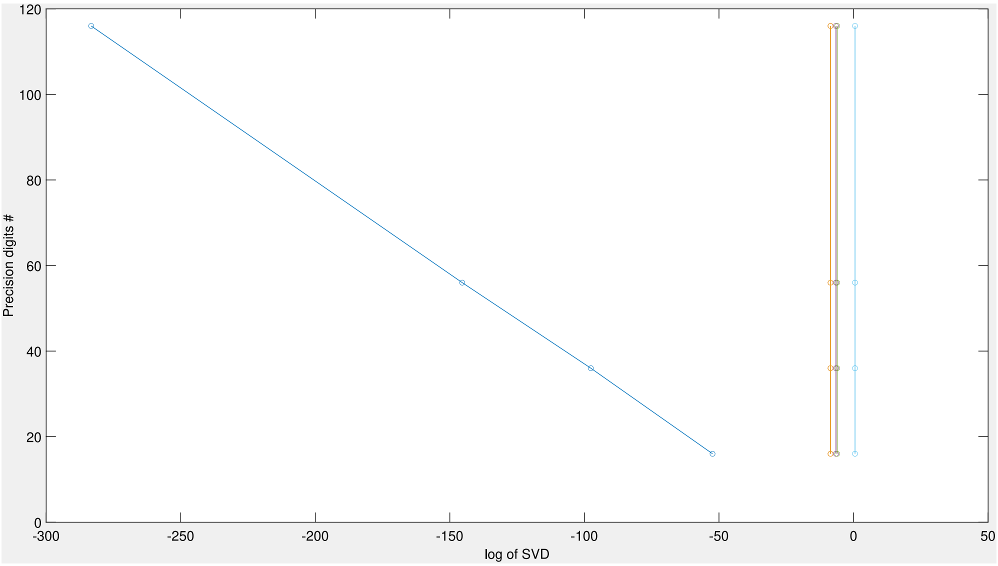


Fig. 3. An example of stabilization, the horizontal axis shows the logarithm of the singular values of a matrix $d\boldsymbol{\Omega}_k$ calculated for different precisions which are indicated in the vertical axis. Each of the four curves corresponds to a SV.

Fig. 3 illustrates that the first singular value (lowest in magnitude) does not stabilize as the precision digits are increased and its value keeps changing to a smaller number (a common trend in this problem) while the remaining singular values are indeed stable. In this example, the first singular value would be set to zero, while the non-zero values for the remaining singular values should be accepted.

In Eq. (12) the vector ϵ_{nd_1,nd_2} encompasses all the sources of errors that affect the singular values. Theoretically, the rank of $d\boldsymbol{\Omega}_{k_{nd_1}}$ would correspond to the number of its non-zero singular values. The existence of ϵ_{nd_1,nd_2} however indicates that even if a singular value of $\mathbf{SV}(d\boldsymbol{\Omega}_k)$ is supposed to be zero, the corresponding value of $\mathbf{SV}_{nd_2}(d\boldsymbol{\Omega}_{k_{nd_1}})$ will be non-zero. As such, to calculate the numerical rank of a matrix one needs to select a threshold such that if $\left|SV_j(d\boldsymbol{\Omega}_k)\right| < tol$ the corresponding Singular Value SV_j is considered as zero and is excluded when calculating the rank. The choice of this tol would require estimating the order of the elements of ϵ_{nd} propagating errors through multiple operations.

Choosing an inappropriately small/large number for tol would result in over/underestimating the rank of $d\boldsymbol{\Omega}_k$. Furthermore, it can be shown [26] that in the general case when adding rows/columns to a matrix the smallest, non-zero, singular values of the occurring matrix have a smaller magnitude than the smaller non-zero singular value of the original matrix. Consequently, different tolerances tol have to be used not only when calculating the rank of $d\boldsymbol{\Omega}_k$ at different steps k, but also when calculating the rank of sub-matrices of $d\boldsymbol{\Omega}_k$ by removing columns.

Consider the singular values $\mathbf{SV}nd_2(d\boldsymbol{\Omega}k_{nd_1})$ computed from two realizations of nd_1, nd_2, denoted by $R_1 : (nd_1 = i_1, nd_2 = j_1)$ and $R_2 : (nd_1 = i_2, nd_2 = j_2)$. An increase in either nd_1 or nd_2 will generally tend to decrease ϵ_{nd_1,nd_2}. In addition, if $i_2 > i_1$, then $\epsilon d\boldsymbol{\Omega}k$ will decrease, and if $j_2 > j_1$, then $\epsilon\mathbf{SV}(d\boldsymbol{\Omega}_k)$ will decrease.

Similar to the concept of stabilization in Operational Modal Analysis [27], we expect that the value of a singular value $SV_j nd_1(d\boldsymbol{\Omega}k_{nd_2})$, which is not significantly affected by $\epsilon nd_1, nd_2$, will not undergo substantial relative changes as nd_1 and nd_1 are increased. On the other hand, if SV_j is expected to be zero, it will be entirely dominated by the corresponding error term $\epsilon_{j\,nd_1,nd_2}$, and therefore, the relative changes resulting from increasing nd_1 and nd_2 will be substantial. This motivates the following algorithm:

Algorithm 2.4.

1. $i = 0$
2. $precisions = [dig_1, dig_2]$
3. $prec2 = [dig_2 + 40, dig_2 + 100]$
4. $i = i + 1$
5. Set nd_1 to the i^{th} element of precisions calculate $d\boldsymbol{\Omega}_{k_{nd_1}}$
6. $j = 0$

   a. $j = j + 1$
   b. Set nd_2 to the j^{th} element of $prec2$ calculate: $S_j = \mathbf{SV}_{nd_2}(d\boldsymbol{\Omega}_{k_{nd_1}})$
   c. if j=1 go to 6a.

7. $SV(d\boldsymbol{\Omega}knd_1)(i) = \begin{cases} S_2(i), & \text{if } \frac{S_2(i)-S_1(i)}{S_2(i)} < 0.01 \\ \text{NaN}, & \text{otherwise} \end{cases}, i = 1, \cdots, \text{length of } S_2$
8. if i=1 go to 4
9. $rank = 0$
10. for $i = 1, \cdots$, length of $SV(d\boldsymbol{\Omega}_{k_{dig_2}})$: if $(SV(d\boldsymbol{\Omega}_{k_{dig_2}})(i) - SV(d\boldsymbol{\Omega}_{k_{dig_1}})(i))/SV(d\boldsymbol{\Omega}_{k_{dig_2}})(i) < 0.01 : rank = rank + 1$

The choice of the 1% threshold for steps 7 and 10 of the above algorithm is arbitrary. In the examples studied different thresholds were used for each of the steps with values up to 100% and made no difference to the observability properties, as the rejected singular values tend to have much larger relative differences. The numerical precision in this paper is controlled using the ADVANPIX toolbox [28] within Matlab 2019b [29].

3. Examples

3.1. Example 1: 11-storey building

In this section the discrete-observability properties of the shear-mass system shown in Fig. 4 are examined. The 11 DOFs of the system are the horizontal displacements of the floors, with x_i corresponding to the displacement of the ith floor as indicated in the figure. The states of the system $x_i, \dot{x}_i$ and the stiffness and damping parameters k_i and c_i are to be identified. An unmeasured input p is applied on the middle floor of the structure, the 7th floor. The piecewise constants occurring from applying ZOH to this input for a sampling-time step Δt are also to be identified. The measurements include the displacements of the bottom and top floor $y = [x_1, x_{11}]^{\mathrm{T}}$. The masses of the system m_i are assumed to be known and for simplicity set to $m_i = 1$, for $i = 1, \dots, 11$.

The system is discretized using the Euler integration described in Eq. (8) or equivalently using $N = 1$ in Eq. (9). The properties of the occurring $d\Omega_k$ are presented in the following Fig. 5, where the kth row corresponds to the kth measurement step, the first 22 columns correspond to the initial conditions of $\mathbf{x}_0$, and are indicated in the Figure within the red rectangle, the subsequent 22 columns correspond to the elements of θ, and are indicated in the Figure within the green rectangle, and columns $44 + (j - 1)$ corresponds to the piecewise constant input p_j. If the state, parameter or input corresponding to column j is found as observable after k measurements steps, this is indicated in the Figure below with a blue marker at the corresponding row k and column j of the matrix which is otherwise left blank.

As it can be seen in Fig. 5, the first measurement step at which all of $\mathbf{x}_0$ and θ become observable is the 37th step, i.e., $N_s = 37$. At that time step, the 30 first piecewise constant inputs: $p_0, \dots, p_{29}$ have also become observable and so $L_{p_i} = 37 - 1 - i =\geq 7$, for $i = 0, \dots, 29$. As suggested by Corollary 2.2 the same value of $L = 7$ applies to all piecewise constants, i.e., $L_{p_i} = 7$ for $i > 29$, and indeed this can be verified from the Figure as e.g., p^{30} becomes observable at measurement step $k = 38$.

It is further important to notice that other expected properties of observability are retained in Fig. 5. For example, as soon as a component of $\mathbf{x}_0$ and θ or a piecewise constant input $p^{(i)}$ becomes observable at a step k, then it is found as observable for all subsequent steps. This is a property which always holds. It was not enforced in Algorithm 2.4 so as to validate the results obtained from the calculated numerical ranks. Had the entries of the observability matrix been calculated with the default double precision and had Matlab's default, double precision, rank algorithm been used, the following Fig. 6 would have been obtained.

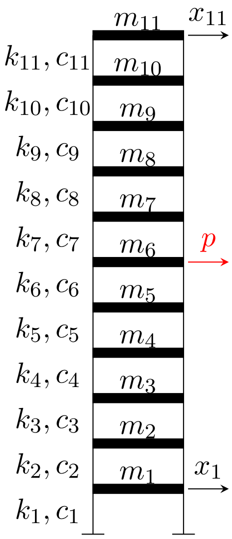


Fig. 4. A 11-storey shear building.

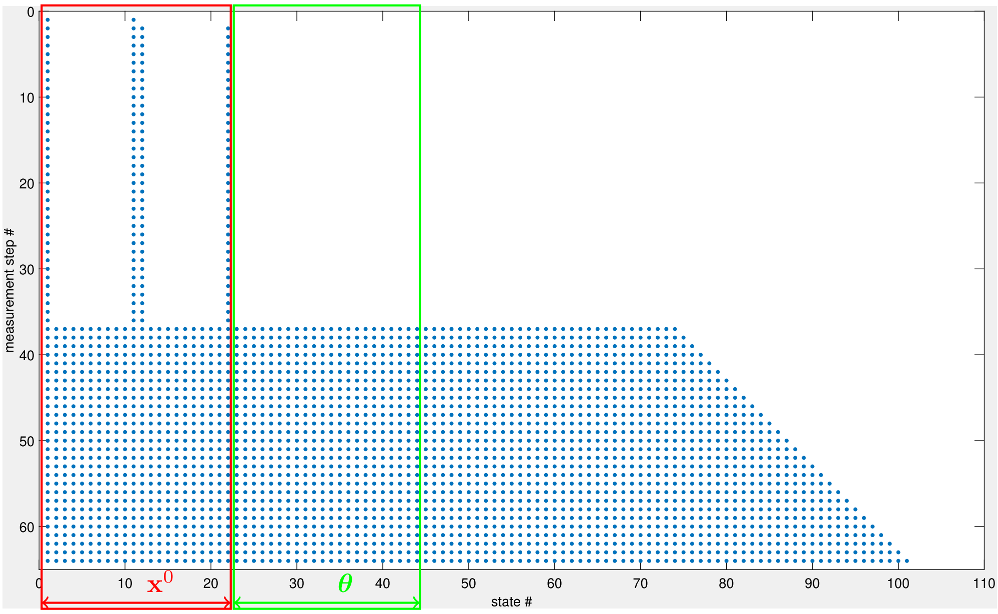


Fig. 5. The Observability matrix of the system shown in Fig. 4 for $N = 1$, rows correspond to different measurement steps, while columns correspond to a different element of the augmented state vector. Empty entries correspond to elements that are k−row unobservable.

The results of Fig. 6 are incorrect and this can easily be inferred simply by the fact that various observability properties are violated, e.g., several states are deemed as observable in a step and then appear as unobservable in subsequent steps. One can further notice that the errors do not have a consistent property. Observable quantities are often deemed as unobservable, but the opposite is also true e.g., as in the 9th step of application of the algorithm where x_4 and x_5 are erroneously deemed as observable. The errors in the calculated observability rank are both due to ϵ_{nd_1} and ϵ_{nd_2}, i.e., the construction of the entries of the Observability matrix using double precision and the subsequent use of the default double precision rank algorithm. It can be further shown that retaining either still leads to erroneous results, thus indicating the need for all the steps in Algorithm 2.4 and the use of higher precision arithmetic.

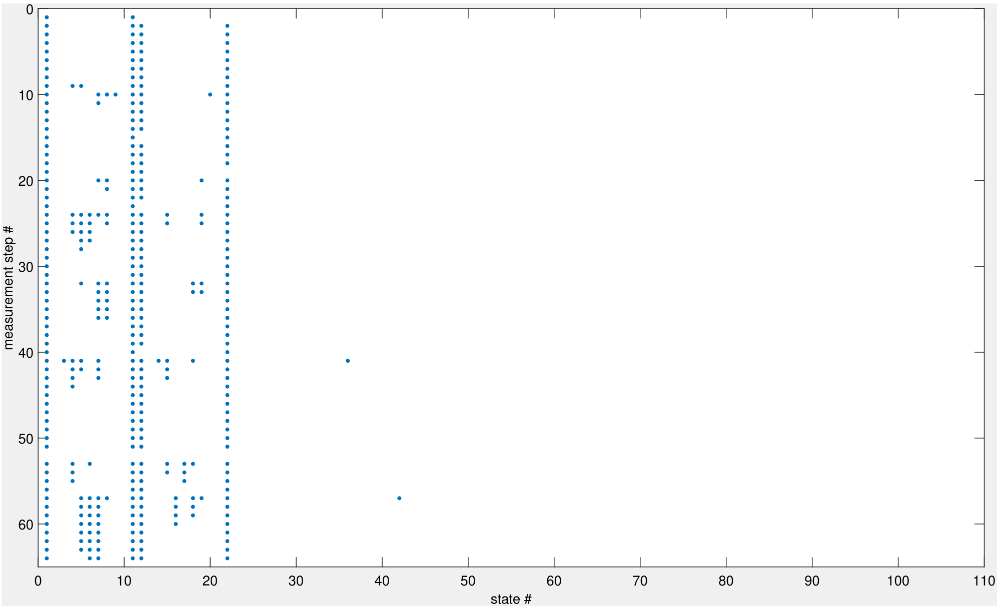


Fig. 6. Repeating the observability analysis of Fig. 5 using double precision.

Table 1
The effect of changing the discretization scheme N on N_s and L.

N	1	2	3	4	5	6	7	11
L	7	4	3	2	2	2	1	1
N_s	37	40	41	42	42	42	43	43

The double precision algorithms used to produce the results of Fig. 6 would approximately correspond to $Nd_1 = Nd_2 = 17$. In contrast, the results of Fig. 5 have been calculated using Algorithm 2.4 for $nd_1 = [90, 100, 110]$ and $nd_2 = [110, 130, 150, 210]$ digits. One should notice that the results in this case are already consistent for simply using the accuracy of 90 digits and perhaps even less than that, but the additional precisions allow confirming that the results have indeed converged. These accuracies are used for all the examples used as they were more than adequate in all test cases.

The continuous observability of the system in this example is also studied using the algorithm suggested in [14], and the results obtained are identical to those presented in Fig. 5 with the following differences in terms of the definitions of the rows and columns of the matrix: While the rows would still correspond to different steps of applying the algorithm, each of those steps would correspond to taking a higher order derivative of the measurements rather than advancing the measurements towards the next time step. The first 44 columns have the same meaning as in the discrete case, i.e., correspond to $\mathbf{x}$ and θ, however, the following columns correspond to higher order time derivatives of the unmeasured input rather than the piecewise constants appearing at a sampling step. One would expect that the observability of the continuous and discrete-time systems are heavily related. It is however worth noticing that the specific discrete-time system for $N = 1$ produces the same structure as the continuous observability matrix, provided the aforementioned correspondence of rows and columns. Hence the continuous system and the discrete-time system for $N = 1$ do not only predict the same observability properties for the various components, but the continuous algorithm would predict the same values for the parameters N_s and L. This is a property which was observed for several example systems that have been tested, but as will be shown later there are cases where the observability matrices produced are not in full agreement. It should also be observed that for the continuous systems the values of N_s and L would not have a directly useful consequence for an identification algorithm as they would conceptually correspond to the number of derivatives of the measurement that have to be included before some of the states, parameters of derivatives of input become observable.

Modifications will be made to this baseline example over the next paragraphs to understand the effect of various parameters on the discrete observability of this system.

3.1.1. The effect of the discretization scheme

For the system of Fig. 5, a set of discrete-time systems are obtained using increasing integer values of parameter N in Eq. (9). It is reminded that larger values of N result in higher-order integration schemes. The discrete observability of those systems, each referred to by the value of N that was used to produce them, is studied using Algorithm 2.4. The system is found to be observable for all the values of N, as such the differences between the different discretization schemes are reflected through the corresponding values for the parameters N_s and L which are reported in the following Table 1.

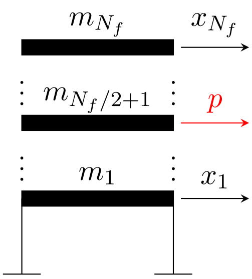


Fig. 7. A shear build with N_f (odd) floors with the displacements of the top and bottom floor being measured and an unmeasured input p applied at the middle floor.

Table 2
The effect of the number of floors N_f on N_s and L.

N_f	3	5	7	9	11	13	15	17	19
L	3	4	5	6	7	8	9	10	11
N_s	9	16	23	30	37	44	51	58	65

From Table 1 it can be seen that generally the values of L decrease or stay the same for any $N_2 > N_1$, i.e., as the discretization scheme becomes of higher order. Similarly, the values of N_s increase or stay the same for any $N_2 > N_1$. Interestingly, for some value of N, 7 in this example, N_s and L reach a maximum and minimum respectively beyond which they do not change. Since theoretically the perfectly time-discretization system would occur for $N \to \infty$, and since for any $N \geq 7$ the system is observable with $N_s = 43$ and $L = 1$ it can be concluded that those are also the properties of the perfectly discretized system.

Using a higher order discretization of the system comes with the disadvantage of the increase of N_s, which means that more time steps are needed before meaningful estimates of $\mathbf{x}$ and θ are obtained. If one looks at the values of N_s in Table 1 the difference is from $N_s = 37$ for $N = 1$, to $N_s = 43$ for $N \geq 7$. For an identification algorithm, this has the implication that the estimates provided in its first $N_s - 1$ steps cannot be separated from their immediate neighbours and hence are not meaningful estimates. However, for a system like the one studied in this example, measurements would typically be obtained for more than 43 time steps, and these additional measurements would allow for the various quantities to be identified becoming observable, i.e., distinguishable from their neighbours, and hence the corresponding estimates from that point and on become meaningful. In contrast, the trends related to L have more significant implications as was discussed earlier.

It should be noted that a different choice of the discretization scheme may have significant effects on the requirements posed by the identification algorithm. For this example, the use of an Euler scheme, i.e., $N = 1$, would not be uncommon but it would require that the identification algorithm involves each piecewise constant input to at least 7 measurement equations. The majority of input estimation algorithms, in a system that does not have direct feedback, would typically involve each piecewise constant to not more than one measurement. Table 1 implies that for such an algorithm, to be able to estimate each of the piecewise constant inputs a time-discretization scheme with $N \geq 7$ would need to be employed. This would crudely correspond to an algorithm of complexity equivalent to a 7th order Runge–Kutta, which would not be a common choice for a system like the one in the example chosen.

Additionally, although systems corresponding to a time discretization with $N = 3$ and $N = 7$ are sufficiently different from a perspective of observability, the numerical differences in their predicted response may be very likely below the accuracy of the sensor and perhaps even below the tolerances of double precision that would typically be used for the time and measurement updating steps of the algorithms. As such while, in theory, an identification algorithm's use of each piecewise constant input in a single measurement can be compensated with a sufficiently high integration scheme, the system may likely be practically unobservable. It instead appears that the use of an identification algorithm that involves each of the piecewise constants to at least $L = 7$ time steps after the one at which the input appears, i.e., the maximum value of L which is required for the case of $N = 1$, would appear as a strategy that is less prone to practical identifiability issues. In that case, regardless of the integration scheme used, the requirements of theoretical observability would be satisfied. This is often important as the user in certain cases has no direct control over the discretization method used.

3.1.2. The effect of the system size

This subsection focuses on the effect of the size of the system on its observability properties with a focus on the values of N_f and L. In the following Fig. 7, a shear building with a total number of floors N_f, with N_f being odd, is shown. The displacements x_1 and x_{N_f}, of the first and last floor respectively, are measured while an unmeasured excitation is applied at the middle floor of the building, i.e., at $m_{(N_f+1)/2}$.

To further isolate the effect of the discretization scheme studied in the previous section, a discretization of $N = 1$ is used for all cases. All systems are observable and as such the results are presented in terms of N_s and L in the following Table 2:

As can be seen in Table 2 for the specific sensor and input setup used, as N_f increases so do the values of L and N_s. The values of the table are described by the following equations: $L = 2 + (N_f - 1)/2$ and $N_s = 2 + (N_f - 1)/2 \times 7$. Although not formally proven, it has been observed that these formulas hold for any of the tested, odd values of N_f. In terms of the predicted values of N_f,

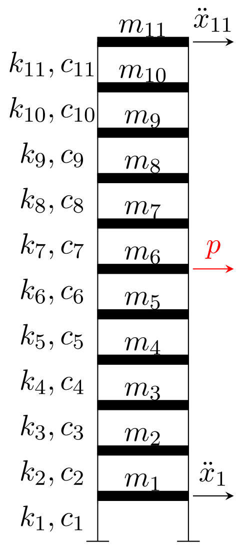


Fig. 8. A shear building with 11 floors with the accelerations of the top and bottom floor being measured and an unmeasured input p applied at the middle floor.

Table 3
Using accelerometers on the 11-storey shear building.

N	1	2	3	4	5	6	7
L	5	3	2	2	1	1	1
N_s	39	41	42	42	43	43	43

the commonly acknowledged problem in the literature of the curse of dimensionality [30] can be observed. The minimum number of measurements before the states and parameters all become observable increases rapidly. As such, a much larger system would require a significant number of steps before any meaningful estimate is obtained. However, the trend described for L is perhaps once again more crucial as it demonstrates how increasing the dimensionality of this system, could lead to significant requirements from an input estimation algorithm in terms of the number of measurements each piecewise constant needs to be involved in.

The comments provided in the previous example regarding the effect of discretization also apply to the systems described in this case. Values of N larger than one would result in the theoretical reduction of L, but as previously discussed this would be difficult to exploit practically.

3.1.3. The effect of sensor placement and type

The accelerometers of Fig. 8 are used as a replacement of the displacement sensors of Fig. 4. The results of this modification are presented in the following Table.

A comparison between Tables 1 and 3 shows that the use of accelerometers has generally the tendency of reducing L and increasing N_s. This is however a much more prominent effect for the Euler discretization, i.e., $N = 1$ and becomes rapidly less important as N increases where regardless of the use of accelerometers or displacements sensors both setups result in $N_s = 43$ and $L = 1$ for a sufficiently large value of N. For the accelerometers, the value of N is 5 but for the displacement sensors, it is $N = 7$. For $N = 1$ this difference of two between accelerometers and displacement sensors can be explained by the fact that in that discretization setup a finite difference description of the acceleration of a floor at a time step t_k, is directly expressed in terms of the corresponding displacements of three successive time steps, e.g., t_k, t_{k+1} and t_{k+2}. Hence, this difference between the acceleration and displacement setups of the values of L and N_s remains consistently equal to 2 regardless of the (odd) number of floors N_f of the shear building. The difference between the two setups is thus much more prominent when the Euler discretization is used and it appears to be of practical exploitation for generally smaller systems.

The previous conclusion however, largely depends on the placement of the sensors (displacement or acceleration) at the first and last floor, with the unmeasured force being applied as far as possible from both sensors, i.e., the middle floor. The location of the sensors versus the unmeasured input also affects the results. In the following Fig. 9, this is investigated for an $N_f = 7$ floors shear system by moving the displacement sensors.

As can be seen in Fig. 9, moving the sensors one additional floor away from the unmeasured excitation increases the value of L by 1 and reduces N_s by 1. The most favourable setup investigated in 9 for reducing L is that shown in Fig. 9(b) with the displacement sensors being measured on the immediately neighbouring floors to the one where the unmeasured excitation is applied. While one could expect to further reduce L by collocating the displacement sensor(s) with the unmeasured excitation at the 4th floor, as Fig.

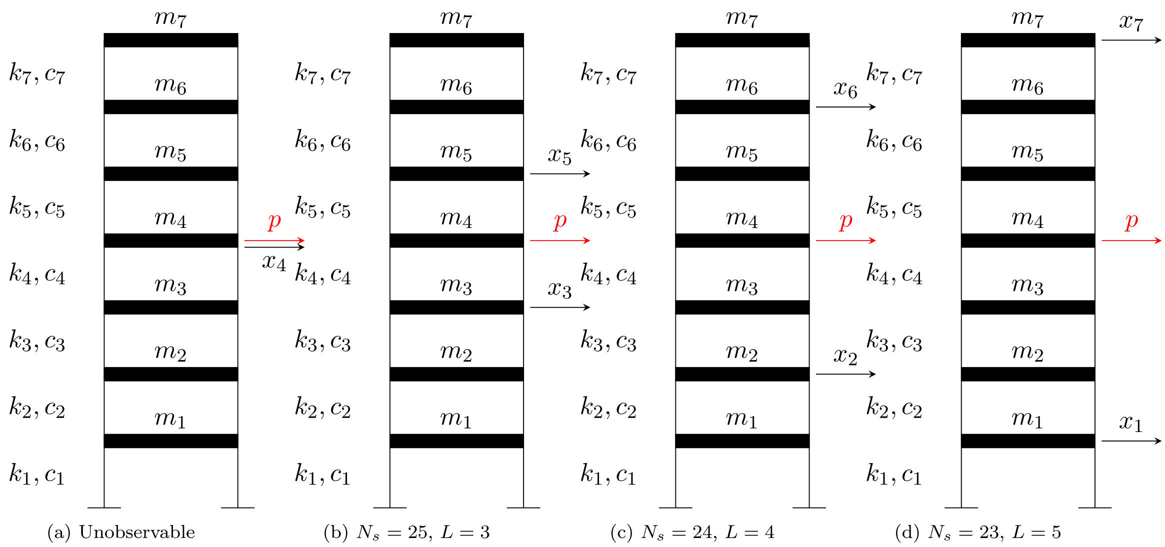


Fig. 9. Investigating the effect of moving the sensors away from the unmeasured input.

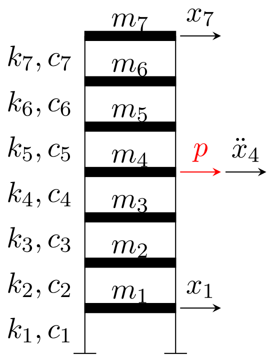


Fig. 10. Collocated accelerometer.

9(a) indicates the collocation of the two displacement sensors on the same floor renders one of them redundant and in this case also renders the system unobservable.

The case of placing the sensors asymmetrically around the unmeasured excitation was also studied where it was found that in the case when one of the displacement sensors is below the unmeasured excitation and the other one above, the values of L and N_s are identical to the symmetric case corresponding to the sensor closer to the unmeasured excitation. As such the optimal strategy for reducing L with displacement sensors would be to place a sensor either on the third/fourth floor and place the other sensor above/below the fourth floor. This would result in $L = 3$. Interestingly, placing a collocated displacement sensor with the input at the fourth floor with only one additional displacement sensor either above or below the fourth floor would result in an unobservable setup.

Based on the previous to reduce L to the value of zero for $N = 1$ one would need to place a collocated accelerometer at the same floor as the unmeasured input, but also place displacement sensors above and below that floor, with one such case for example shown in the following Fig. 10.

It is interesting to note that the setup of Fig. 10 requires one more sensor in comparison to Fig. 9(d). This extra accelerometer is not necessary for making the system observable, it is instead only beneficial for decreasing the value of L to zero. It should further be noticed that while it would appear 'reasonable' that the system would be observable after omitting one of the two displacement sensors, as discussed earlier that would be an unobservable system.

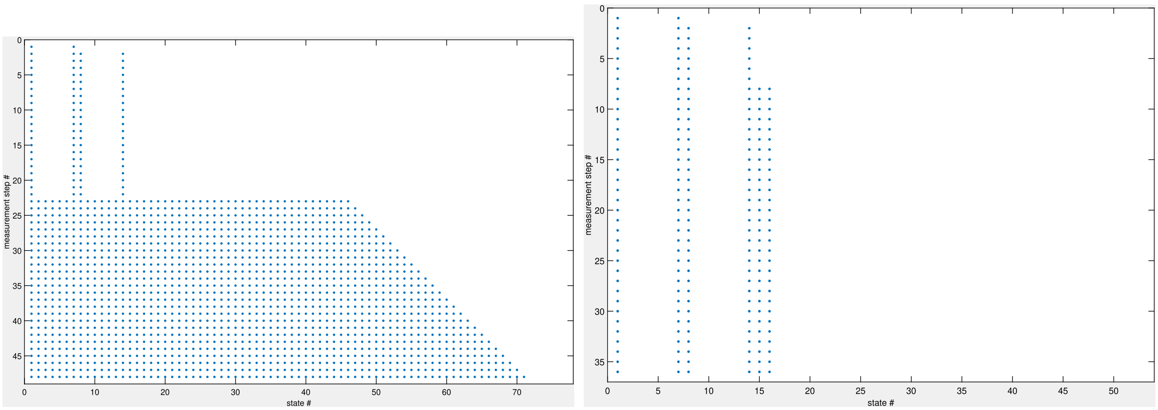
This last result is perhaps reasonable although not straightforward to see without running the observability analysis of the system first. The next example will focus on a case that is even more counter-intuitive.

3.2. Example 2: A special case leading to lack of observability

In this example a case where a commonly employed simplification leads to deteriorating the observability of the system is demonstrated. In the following Fig. 11, a 7-storey shear system is shown with the masses known, the displacements of the first

(a) (b)

Fig. 11. A shear building where (a) the different k_i and c_i for each floor are to be identified (b) to common tog all floors k and c are to be identified.



(a) (b)

Fig. 12. The Observability matrix for the cases of (a) Sub Fig. 11(a) and (b) (a) Sub Fig. 11(b).

and last floor measured, an unmeasured force p applied to the 4th floor. The case shown in Sub Fig. 11(a) is compatible with the cases studied in the previous Examples, with each floor i having a different stiffness k_i and coefficient of damping c_i which are all to be identified. Often that turns out to be challenging in practice, and so it is commonly assumed that an easier case for a system to be identified would be that of having all the floors have the same parameters k and c as shown in the Sub Fig. 11(b). Especially in the case where simulated data are used, researchers would often start from the simplified system in Sub Fig. 11(b), before testing the seemingly more complex case of Sub Fig. 11(b). The Euler integration scheme, $N = 1$ is used to convert both systems to discrete-time.

Similarly to what was done in example 1 for Fig. 5, the properties of the occurring $d\Omega_k$ are presented in the following Fig. 12, where the kth corresponds to the kth measurement step, the first 14 columns correspond to the initial conditions of $\mathbf{x}_0$, the subsequent 14 columns for the case of Sub Fig. 12(a) and 2 columns for the case of Sub Fig. 12(b) correspond to the elements of $\boldsymbol{\theta}$, and the remaining columns correspond to the piecewise constants of the unmeasured input p.

It can be seen in Fig. 12(a) that for the system described in Sub Fig. 11(a) all the states parameters and inputs become observable. However, this is not true for its simplified version in Fig. 12(b). Although the number of unknowns has been reduced drastically, through the condition that all floors have the same properties, this has also resulted in rendering all the piecewise constants of the applied input, and the majority of the dynamic states unobservable. The only observable quantities for that case are the displacements and velocities of the first and last floor, which was expected as the displacements of those floors are measured, and the parameters k and c.

This last point is interesting, the intuition would guide a researcher to construct this simpler cases for an 'easier' estimation of those parameters, but a counter-intuitive result is that this comes at the price of losing observability for many other important quantities of the problem. From a first analysis, it would also appear that the observability properties shown in Figs. 12(a) and 12(b) are contradictory. If the system in 11(a) is observable and all parameters are observable for arbitrary values of k_i and c_i, then one would expect that this would also be the case for the special case where the $k_i = k$ and $c_i = c$ $\forall i$. It should be reminded however, that there are realizations of the states parameters and inputs where the observability matrices lose rank as also discussed in [13]. Typically those values are not associated to a specific physical property, however in the case of this systems such a singularity occurs

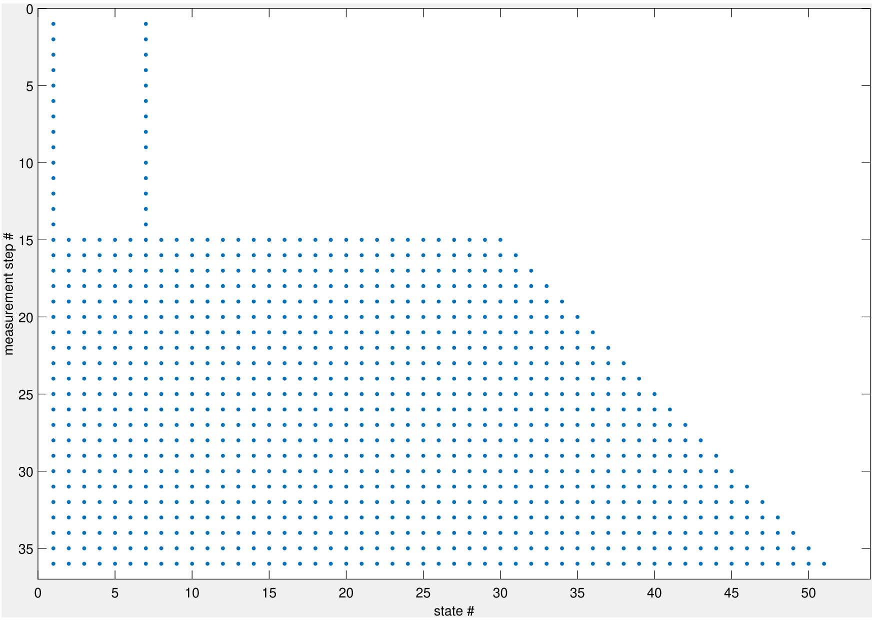


Fig. 13. The observability of the system in Fig. 11(b) for $N = 5$.

for the physically meaningful case of the parameters of all floors being identical. As such, while certainly being counter-intuitive, the two results are not contradictory.

It should also be mentioned that the discrete observability properties that occur for the two systems shown in Fig. 11, both of which are discretized with $N = 1$, are in agreement with the continuous observability properties of the two systems as investigated using the algorithm suggested in [14].

Furthermore, as in both systems the parameters are observable and hence could eventually be identified, assuming them to be known should not change the observability of the states and inputs of the system. To demonstrate this, the observability of the systems in Fig. 11 under the assumption that all their parameters are known is investigated. In this case this becomes the state and input observability problem of a continuous linear system of the form:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_{\mathrm{u}}\mathbf{u} + \mathbf{B}_{\mathrm{p}}\mathbf{p} \tag{13}$$

$$\mathbf{y} = \mathbf{G}\mathbf{x} + \mathbf{J}_{\mathrm{u}}\mathbf{u} + \mathbf{J}_{\mathrm{p}}\mathbf{p} \tag{14}$$

$\mathbf{A}$ being the state-space matrix of the system, $\mathbf{B}_{\mathrm{p}} = [\mathbf{0}^{1*10}, 1, \mathbf{0}^{1*3}]^T$, $\mathbf{G} = \begin{bmatrix} 1 & \mathbf{0}^{1*5} & 0 & \mathbf{0}^{1*7} \\ 0 & \mathbf{0}^{1*5} & 1 & \mathbf{0}^{1*7} \end{bmatrix}$, $J_{\mathrm{p}} = 0$.

As stated in [13] the continuous observability of such a system would be given by the following matrix:

$$d\Omega_k = \begin{bmatrix} \mathbf{G} & \mathbf{J}_{\mathrm{p}} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{GA} & \mathbf{GB}_{\mathrm{p}} & \mathbf{J}_{\mathrm{p}} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{GA}^2 & \mathbf{GAB}_{\mathrm{p}} & \mathbf{GB}_{\mathrm{p}} & \mathbf{J}_{\mathrm{p}} & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ \mathbf{GA}^k & \mathbf{GA}^{k-1}\mathbf{B}_{\mathrm{p}} & \mathbf{GA}^{k-2}\mathbf{B}_{\mathrm{p}} & \mathbf{GA}^{k-3}\mathbf{B}_{\mathrm{p}} & \cdots & \mathbf{J}_{\mathrm{p}} \end{bmatrix} \tag{15}$$

The rank of this observability matrix can be investigated using symbolic numbers where one can again verify that for the state and input estimation of the two systems under study in Fig. 11, the observability properties are retained: all the states and input and its derivatives would be observable for the more complicated system in Sub Fig. 11(a), and only the displacement and velocities of the measured floors are observable for the seemingly simpler system in Sub Fig. 11(b). As in the previous case, this counter-intuitive result is once again a consequence of the more general observability matrix obtained from the system in 11(a) losing rank for the special case of $k_i = k$ and $c_i = c$ $\forall i$. This property is retained also in the case where the parameters are to be estimated. It hence appears that this counter-intuitive behaviour is generated due to the inputs being unmeasured, while in systems where the inputs are known or measured it is rarely to observe situations where similar simplifying assumptions on the parameters would result in loss of observability.

So far within this example the system has been discretized with $N = 1$. It was also shown that the results are in agreement with the corresponding results of continuous observability. While a change of N in other examples has been shown to have an effect on N_s and L it did not appear to ultimately change the observability properties of the states, parameters or inputs. The corresponding matrix for $N = 5$ is shown in the following Fig. 13.

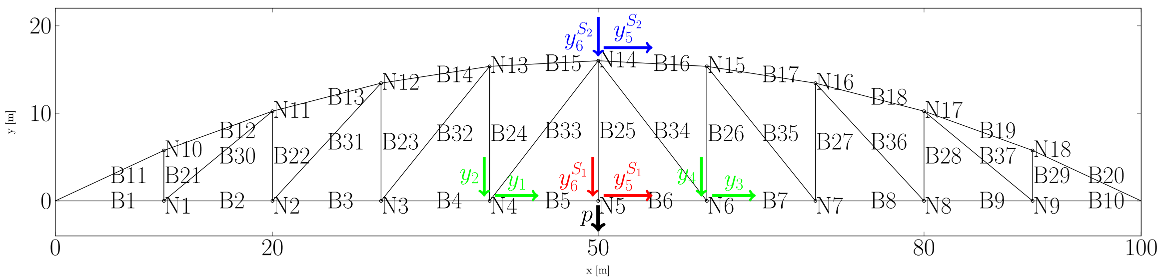


Fig. 14. A truss bridge. Each of the nodes N_i has two DOFs: horizontal $x_{(i-1)2+1}$ and vertical $x_{(i-1)2+2}$. An unmeasured load p is applied at node N_5. Sensor setup S1: uses the displacement sensors $(y_1,\dots,y_4,y_5^{S_1},y_6^{S_1})$ and S2: $(y_1,\dots,y_4,y_5^{S_2},y_6^{S_2})$.

This observability matrix obtained for the system in 11(b) for $N = 5$, shown, is different to the one that was shown for the same system for $N = 1$ in Fig. 12(b). It can be seen that increase of N resulted in making the states and inputs of the system observable. This result remains the same as N is further increased and can hence be treated as indicative of the observability properties of the perfectly discretized system. This is hence a case where: (i) the observability of several dynamic states and the piecewise constants related to the unmeasured inputs depends on the discretization scheme used, and (ii) the continuous observability of the system is not in agreement with the discrete observability (of the perfectly discretized system).

This example demonstrates that it is worth investigating the observability of systems with unmeasured inputs rather than making a guess based on the observability of similar systems with known or measured inputs. It further demonstrates that, though it is not typically the case, the observability of a discrete-time system can be different from the continuous system and also depends on the time integration scheme used.

3.3. Example 3: Truss bridge

This example investigates a larger structure that involves a larger number of states and parameters to be identified. A truss type of bridge is shown in the following Fig. 14. The bridge is pinned at its two ends. Its overall span is 100 [m], and its maximum height is 16 [m] at node N_{14}. The remaining N_i $i = 1,\dots,18$ each has two DOFs a horizontal and a vertical displacement, $x_{(i-1)2+1}$ and vertical $x_{(i-1)2+2}$ respectively. The time derivatives of those displacements $\dot{x}_i$ comprise the remaining dynamic states of the system x_t.

The bridge is modelled using truss elements B_i, $i = 1,\dots,37$, with the stiffness of each of those elements k_i to be identified. The mass of the bridge is assumed to be known and it is further assumed that the mass is concentrated at the nodes. The damping matrix of the structure CC is constructed assuming Rayleigh damping with $[CC] = \alpha[M] + \beta[K]$, where $[M]$ and $[K]$ are the mass and stiffness matrices of the structure, and the scalar parameters α and β are also to be identified.

An unmeasured vertical load p is applied at the midspan of the deck of the bridge at node N_5. In total 72 dynamic states are to be identified, together with 39 time-invariant parameters, and the piecewise constants of the unmeasured input p_i. Fig. 14 investigates two different setups. Both sensor setups use as the first four measurements the displacements of nodes N_4 and N_6. In the first sensor setup, S_1, the two additional measurements are the displacements at the node where the load is applied, N_5. In contrast, S_2 uses as the two displacements of the node above the load, N_{14} as its fifth and sixth sensors.

For the time-discretization, $N = 1$ is used. The results show that only S_1 is fully observable. For S_1, $N_s = 22$ and $L = 20$. While S_2 does not become observable the observability properties as indicated by the following Figure are of interest.

As can be seen in Fig. 15, the 9th and 46th states remain unobservable, i.e., the vertical displacement and velocity of node N_5 where the unmeasured load is applied. Additionally, all the piecewise constants related to the unmeasured input remain unobservable. Despite that, significant information is provided by the setup: all other dynamic states are observable and all the parameters $\boldsymbol{\theta}$ of the problem, consisting of k_i, α and β, are also observable. As such all, the structural parameters of the problem are observable as is also the case for most dynamic states. Hence, if one was to directly identify such a system using this setup, one would obtain a wealth of information that would provide confidence in the results. This may however, lead to missing the fact that the estimated values for at least x_9, $\dot{x}_9$ and the unmeasured input at various time instances $p_i^{(1)}$ cannot be good and may very well lead to erroneous conclusions.

3.4. Example 4: Data fusion

This example will make use of a system with geometric non-linearities and with multiple unmeasured inputs acting on the system. As shown in the following Fig. 16, a rigid body is monitored with accelerometers located at the four corners of the body and with a camera. The camera monitors the horizontal and vertical position of four targets P_i, with $i = 1,\dots,4$, on the body. Due to tracking errors the coordinates of the targets $[x_{P1},\dots,x_{P4},y_{P1},\dots,y_{P4}]$ with respect to the frame fixed on the lower right corner of the body defined by the unit vectors e_x and e_y are to be identified. The accelerometers are fixed on the body with $a_{1h},\dots,a_{4h}$ parallel to e_x and a_{1v} parallel to e_y.

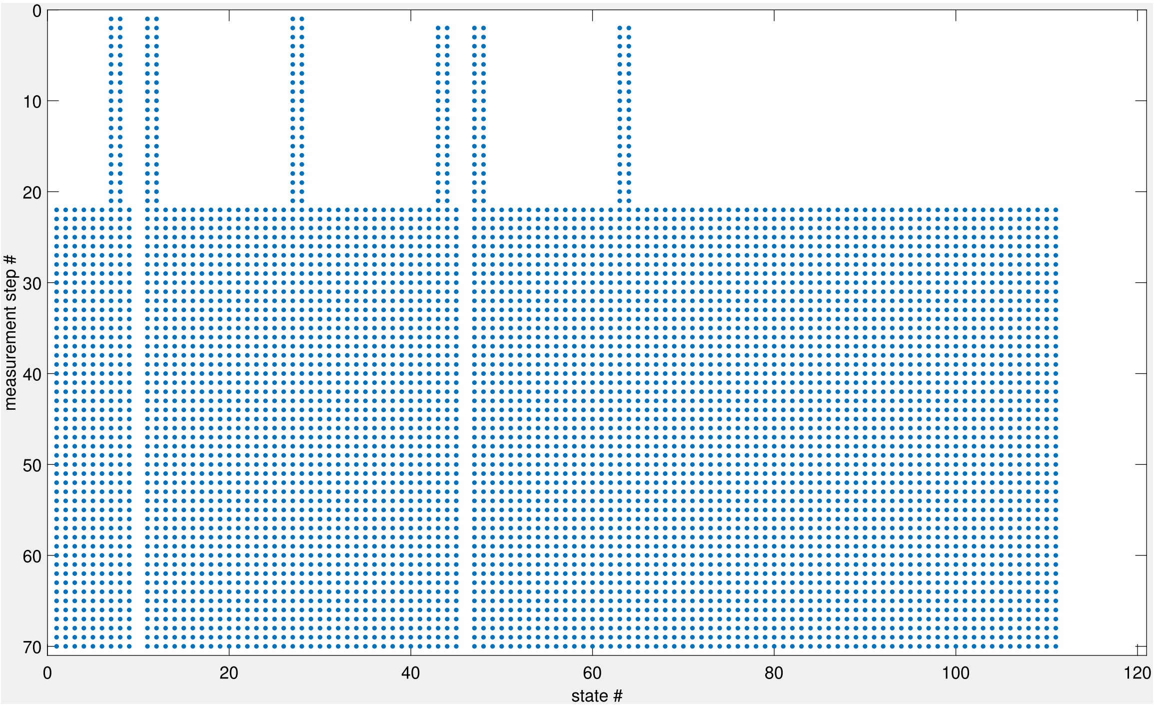


Fig. 15. The observability matrix for sensor setup S_2 of the system shown in Fig. 14.

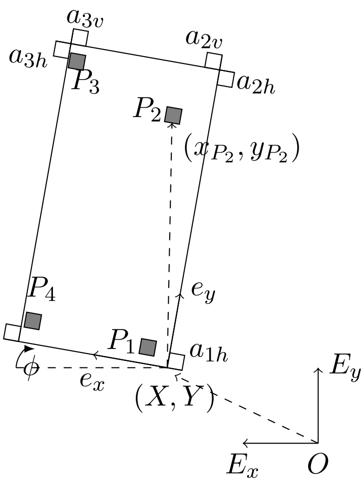


Fig. 16. A rigid body monitored with a set of accelerometers and a camera tracking targets.

The movement of the rigid body is expressed in terms of the location of the bottom right corner (X, Y) with respect to a non-moving frame defined by point O and the non-rotating unit vectors E_x and E_y, and the rotation ϕ.

Unlike a typical data fusion formulation the joint information provided by the accelerometers and the camera will be used for not only updating the displacements, rotations and corresponding velocities of the rigid body but also the accelerations of the rigid body $\ddot{X}$, $\ddot{Y}$ and $\ddot{\phi}$.

The dynamic states of the system are $\mathbf{x} = [X, Y, \theta, \dot{X}, \dot{Y}, \dot{\phi}]$ and the 8 parameters:

$[x_{P1}, \dots, x_{P4}, y_{P1}, \dots, y_{P4}]$.

The system has three independent inputs $p^{(1)}, p^{(2)}, p^{(3)} = [\ddot{X}, \ddot{Y}, \ddot{\phi}]$, which are discretized in time using ZOH with the corresponding piecewise constants at the different time steps to be identified.

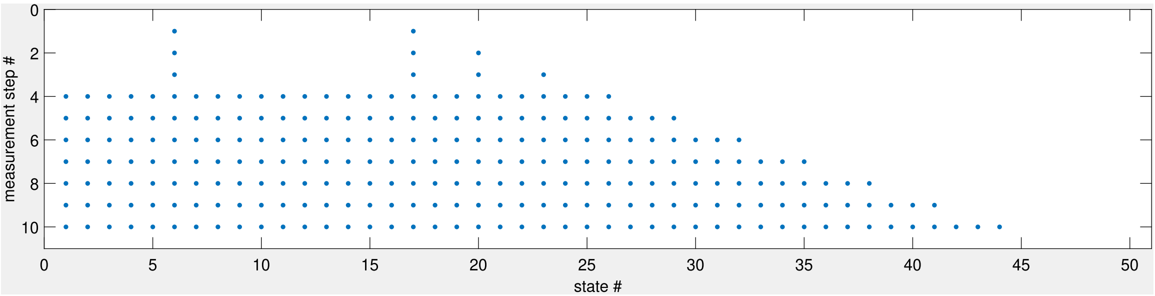


Fig. 17. Observability matrix for the system and measurements presented in Fig. 16.

The state-space equations of the dynamic states are:

$$\begin{aligned}
\dot{x}_1 &= \dot{X}\\
\dot{x}_2 &= \dot{Y}\\
\dot{x}_3 &= \dot{\phi}\\
\dot{x}_4 &= p^1\\
\dot{x}_5 &= p^2\\
\dot{x}_6 &= p^3
\end{aligned} \tag{16}$$

The measurements of the camera for the horizontal and vertical coordinates of a target P_i can be expressed as:

$$\begin{aligned}
X_{P_i} &= X + x_{P_i} \cos(\phi) - y_{P_i} \sin(\phi)\\
&= x_1 + \theta_{(i-1)\,2+1} \cos\left(x_3\right) - \theta_{(i-1)\,2+2} \sin\left(x_3\right)\\
Y_{P_i} &= Y + x_{P_i} \sin(\phi) + y_{P_i} \cos(\phi)\\
&= x_2 + \theta_{(i-1)\,2+1} \sin\left(x_3\right) + \theta_{(i-1)\,2+2} \cos\left(x_3\right)
\end{aligned} \tag{17}$$

The measurement equations for the accelerometers are given by:

$$\begin{aligned}
a_{ih} &= -x_{c_i}\dot{\phi}^2 - \ddot{\phi}\, y_{c_i} + \ddot{X}\cos(\phi) + \ddot{Y}\sin(\phi)\\
&= x_{c_i} {x_6}^2 - p^{(3)} y_{c_i} + p^{(1)}\cos(x_3) + p^{(2)}\sin(x_3)\\
a_{iv} &= -y_{c_i}\dot{\phi}^2 + p^{(3)} x_{c_i} - p^{(1)}\sin(\phi) + p^{(2)}\cos(\phi)
\end{aligned} \tag{18}$$

where x_{c_i} and y_{c_i} are the coordinates of corner i in the frame fixed on the body which are known and not to be estimated, hence they are not included in the vector of unknown parameters θ.

As can be seen in Fig. 17 all states, parameters and inputs become observable. This means that this data fusion approach would provide not only a fused estimate of the displacements and velocities of the rigid body but also its accelerations. Additionally, it would do so despite the lack of the exact knowledge of the position of the camera targets on the body, under the assumption however that those would not change with respect to the frame fixed on the body. Additionally, the measurement step at which all states, parameters and some of the piecewise constants of the three inputs become observable is the 4th, i.e., $N_s = 4$. It is further interesting to observe that $\dot{\phi}_0$ and $u^{(3)}_0$ are observable from the very first measurement, at t_0. Even though $p^{(2)}$ and $p^{(1)}$ become observable for the first time at the 4th measurement step it is worth noting that $L_{p^{(1)}} = L_{p^{(2)}} = L_{p^{(3)}} = 0 = L$. In other words, all piecewise constant inputs appearing from t_3 and can be identified without any lag. This is a very useful property as it indicates that this data fusion problem can provide real-time information about the accelerations of the system.

4. Conclusions

This paper introduced a new algorithm to assess the observability of discrete-time systems with unmeasured inputs that have been sampled using a zero-order-hold scheme. For such systems the kth step of the algorithm is associated to stepping forward in time and obtaining the measurements obtained at a new time step. This allows for not only assessing the observability of a state, parameter or piecewise constant related to an unmeasured input, but also the time step at which they have become observable. This leads to the introduction of two new properties for such systems: the time step at which all the states, parameters and some of the piecewise constant inputs become observable, N_s, and the number of time steps between the appearance of a piecewise constant input $p_j^{(i)}$ and it becoming observable $L_{p_j^{(i)}}$. These parameters are important for identification purposes, especially because they imply the minimum number of time-steps whose measurements will need to be considered before being able to properly estimate the quantity. Especially the requirements implied by L pose challenges for input estimation algorithms.

The paper also introduces a regression scheme for estimating the entries of the discrete observability matrix and pairs it with an algorithm that assesses the rank of the matrix using variable precision arithmetic. This combination allows for the investigation of larger engineering systems than those that can be assessed by the symbolic observability algorithms used for continuous observability [13]. Such systems would typically be made discrete in time using an integration scheme of choice and the paper incorporates the effect of this time integration to various properties of the observability matrix.

The examples chosen demonstrate the use of the algorithm on a set of representative problems which are commonly studied by state-parameter-input estimation algorithms. All examples demonstrate that the integration method used has a significant effect on N_s and L of the studied system. It is worth noting that the very commonly employed Euler discretization, for $N = 1$, results in larger values of L and hence more stringent requirements for the input estimation algorithms. It is also shown that the sensor placement has an effect on N_s and L and that this effect is increasingly more important for smaller values of N. The relevant examples also indicated that a commonly assumed requirement for certain algorithms, that of physical collocation of an accelerometer to the unmeasured input force, is not a necessity. It can, at least in theory, be circumvented through the use of higher-order integration schemes (larger values of N), and instead typically a requirement when the Euler integration is used.

The third example discusses the case of the observability of a truss bridge, demonstrating how the algorithm can be used in a realistic application of practical interest. The fourth example demonstrates the use of the algorithm for a geometrically non-linear problem: that of data-fusion of accelerometers and camera measurements, demonstrating that it is possible to jointly improve the estimated displacements and accelerations of the system.

The most counter-intuitive example is the second example where it is shown that simplifying assumptions on the model can result in loss of its observability, while further indicating that there are situations where the chosen integration scheme, expressed through N, can also change the observability of some of the quantities of the system. This is also the only example where the authors have observed that the results of the continuous observability of the system do not predict the corresponding quantities of a system discretized with the Euler integration scheme.

All the examples demonstrate that the discrete observability investigation can offer valuable intuition regarding which models can properly be estimated under the presence of unmeasured inputs. The predicted values of N_s and especially L have to be taken into account when designing or using input(-state-parameter) estimation algorithms.

A follow-up paper of this work will demonstrate how the predicted values of the observability algorithm, and especially L, can be factored in relevant algorithms to allow for proper estimation of the relevant parameters. The parametrization of the inputs using a zero-order-hold has been used since it is commonly used in input estimation algorithms. However, the concepts presented in this work can be tailored to other parametrizations of the inputs.

CRediT authorship contribution statement

M.N. Chatzis: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **K. Maes:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **G. Lombaert:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The first Author would like to acknowledge the EPSRC OTG, United Kingdom EP/W001098/1 and the KUL Network Fellowship, Belgium for the financial support during his visit to KUL.

Data availability

Data will be made available on request.

References

[1] L. Ljung, System identification, in: Signal Analysis and Prediction, Springer, 1998, pp. 163–173.
[2] C.R. Farrar, K. Worden, An introduction to structural health monitoring, Philos. Trans. R. Soc. A: Math. Phys. Eng. Sci. 365 (1851) (2007) 303–315.
[3] E. Lourens, E. Reynders, G. De Roeck, G. Degrande, G. Lombaert, An augmented Kalman filter for force identification in structural dynamics, Mech. Syst. Signal Process. 27 (2012) 446–460.
[4] V. Dertimanis, E. Chatzi, S. Eftekhar Azam, C. Papadimitriou, Input-state-parameter estimation of structural systems from limited output information, Mech. Syst. Signal Process. 126 (2019) 711–746.
[5] K. Maes, F. Karlsson, G. Lombaert, Tracking of inputs, states and parameters of linear structural dynamic systems, Mech. Syst. Signal Process. 130 (2019) 755–775.
[6] R. Hermann, A. Krener, Nonlinear controllability and observability, IEEE Trans. Autom. Control 22 (1977) 728–740.
[7] S. Diop, M. Fliess, Nonlinear observability, identifiability, and persistent trajectories, in: Proceedings of the 30th IEEE Conference on Decision and Control, 1991, pp. 714–719 vol.1.

[8] M. Chatzis, E. Chatzi, A. Smyth, On the observability and identifiability of nonlinear structural and mechanical systems, Struct. Control. Heal. Monit. 22 (2015) 574–593.
[9] K. Maes, M. Chatzis, R. Vandebril, G. Lombaert, Observability of modally reduced order models with unknown parameters, Mech. Syst. Signal Process. 146 (2021) 106993.
[10] X. Shi, M. Williams, M. Chatzis, A robust algorithm to test the observability of large linear systems with unknown parameters, Mech. Syst. Signal Process. 157 (2021) 107633.
[11] A. Sedoglavic, A probabilistic algorithm to test local algebraic observability in polynomial time, J. Symbolic Comput. 33 (2002) 735–755.
[12] A. Martinelli, Extension of the observability rank condition to nonlinear systems driven by unknown inputs, in: Proceedings of the 23rd Mediterranean Conference on Control and Automation, MED 2015, Torremolinos, Spain.
[13] K. Maes, M. Chatzis, G. Lombaert, Observability of nonlinear systems with unmeasured inputs, Mech. Syst. Signal Process. 130 (2019) 378–394.
[14] X. Shi, M. Chatzis, An efficient algorithm to test the observability of rational nonlinear systems with unmeasured inputs, Mech. Syst. Signal Process. 165 (2022) 108345.
[15] A.F. Villaverde, N. Tsiantis, J.R. Banga, Full observability and estimation of unknown inputs, states and parameters of nonlinear biological models, J. R. Soc. Interface 16 (156) (2019) 20190043.
[16] A. Martinelli, Nonlinear unknown input observability and unknown input reconstruction: The general analytical solution, Inf. Fusion 85 (2022) 23–51.
[17] X. Shi, M. Chatzis, Lie symmetries of nonlinear systems with unknown inputs, Mech. Syst. Signal Process. 188 (2023) 110027.
[18] K. Rieger, K. Schlacher, J. Holl, On the observability of discrete-time dynamic systems—A geometric approach, Automatica 44 (8) (2008) 2057–2062.
[19] M. Witczak, V. Puig, D. Rotondo, P. Witczak, A necessary and sufficient condition for total observability of discrete-time linear time-varying systems, IFAC-Pap. 50 (1) (2017) 729–734.
[20] M. Aoki, On observability of stochastic discrete-time dynamic systems, J. Franklin Inst. 286 (1) (1968) 36–58.
[21] L. Shen, J. Sun, Q. Wu, Observability and detectability of discrete-time stochastic systems with Markovian jump, Systems Control Lett. 62 (1) (2013) 37–42.
[22] L. Moysis, N. Karampetakis, E. Antoniou, Observability of linear discrete-time systems of algebraic and difference equations, Internat. J. Control 92 (2) (2019) 339–355.
[23] M. Spivak, Calculus on Manifolds: A Modern Approach to Classical Theorems of Advanced Calculus, Avalon Publishing, 1965.
[24] M.T. Heath, Scientific computing: an introductory survey, revised, SIAM, 2018.
[25] B.D. Jager, The use of symbolic computation in nonlinear control: Is it viable? IEEE Trans. Autom. Control 40 (1995).
[26] A. Pavel, (https://math.stackexchange.com/users/90996/algebraic-pavel). Changes in singular values of matrix when rows are added. Mathematics Stack Exchange. https://math.stackexchange.com/q/1198556 (version: 2015-03-20).
[27] B. Peeters, G. De Roeck, Stochastic System Identification for Operational Modal Analysis: A Review , J. Dyn. Syst. Meas. Control. 123 (4) (2001) 659–667.
[28] M. Advanpix, Multiprecision computing toolbox, Advanpix Tokyo, 2022.
[29] T.M. Inc., MATLAB version: R2019b, The MathWorks Inc., Natick, Massachusetts, United States, 2019.
[30] R. Bellman, Dynamic programming, princeton univ, Press Princet. N. J. 39 (1957).